A Theory of Firm Boundaries and Knowledge Sharing*

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Abstract

We study how asset ownership affects collaborations where parties need to share private knowledge but are vulnerable to non-contractible expropriation. Ownership can protect the owner's knowledge by reorganizing production so that certain parties need not be informed, or by ensuring that some parties who must be informed lack the means to expropriate. However, because ownership also gives the owner the ability to expropriate, others may withhold their knowledge from the owner. We apply the theory to understand how knowledge hubs, vertical spin-offs, and acquisitions shape the use of knowledge.

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1 Introduction

Firm boundaries are often drawn to protect proprietary knowledge (Teece, 1980; Liebeskind, 1996). Consistent with this view, recent empirical work highlights acquisitions as a mechanism for facilitating knowledge transfers (Demirer and Karaduman, 2025) and suggests that such transfers may be a primary purpose of ownership (Atalay, Hortaçsu, and Syverson, 2014). This paper studies when and why firm boundaries shape the transfer and use of knowledge.

As a motivating example, consider General Motors' (GM) decision to spin off its parts division, Delphi Automotive, in 1999. We argue that integration protected GM's knowledge but discouraged rival automakers from sharing proprietary knowledge with Delphi for fear that GM would get access to it.[1] Spinning off Delphi was a way to make it a more attractive supplier to GM's rivals, but only by giving up some of the knowledge protection that integration gave GM.

Our argument builds on three premises. First, the use of knowledge is *non-contractible*: once knowledge is shared, it cannot be taken back, and it is difficult to restrict how the recipient uses it. The recipient may expropriate it for private gain at the expense of the original holder, for example by applying it to competing products or developing competing follow-on technologies. This feature of knowledge distinguishes it from tangible inputs (Arrow, 1962).

Second, knowledge must be combined with a productive asset such as a specialized plant or equipment. Although parties cannot contract on how knowledge is used, they can allocate ownership of the asset, and hence the *control rights* that govern its use. These control rights can protect the owner's knowledge in two ways: (i) they allow the owner to apply its knowledge directly without disclosing it, and (ii) when disclosure is necessary, they allow the owner to prevent a partner from using the asset to engage in knowledge expropriation.

Finally, ownership confers *residual* rights of control associated with the asset — that is, *all* rights not specified in a contract (Grossman and Hart, 1986). This set of rights is a bundle: some of these rights protect the owner, while others expose non-owners to expropriation. In the GM–Delphi example, GM's ownership allowed it to prevent expropriation by Delphi, but it also gave GM control over production decisions related to other automakers that relied on Delphi. As a result, serving a rival required GM to receive the rival's proprietary knowledge and, if informed, GM could expropriate it.

We develop a model in which parties collaborate on projects. Each project specifies a collection of tasks. A task consists of an action and some target party's privately observed state and is completed if the action matches that state. Projects exhibit O-ring

[1] Delphi noted in its initial public offering prospectus: "[W]e believe that many major VMs [vehicle manufacturers] remain reluctant to source from us because they fear that GM might obtain access through us to confidential information regarding their vehicle designs and manufacturing processes" (Delphi Automotive Systems Corporation, 1999).

complementarities (Kremer, 1993): a project succeeds and generates value only if all tasks are completed. Performing these tasks may require parties to receive information about others' states. But if they are sufficiently informed, they may attempt to expropriate that information. A single productive asset is required for the completion of certain tasks, and ownership confers control over the actions involved in those tasks. Some parties may also need the asset to engage in expropriation, and ownership allows them to do so.

We begin by analyzing this setting through the lens of information design (Bergemann and Morris, 2019). Given an ownership structure, an omniscient mediator chooses a decision rule that maps states into private action recommendations. The mediator's goal is to maximize total surplus. The information conveyed by the recommendations determines how precisely parties learn about others' states. This approach establishes a benchmark that isolates how the non-contractible use of knowledge limits total surplus.

Our main analysis focuses on the case in which expropriation is sufficiently costly that it is optimal to deter it: a party capable of expropriation must not receive enough information about another party's state to make expropriation profitable. For each task, this requirement places an upper bound on how precisely information about the target's state can be disclosed to the acting party. We refer to this upper bound as the task's *safety threshold*.

Our first main result characterizes an optimal decision rule for any fixed ownership structure. Whenever this decision rule recommends a correct action for a low-threshold ("hard") task, it also recommends correct actions for all higher-threshold ("easier") tasks. This nested property is optimal because tasks are complementary within a project, so additional precision on an easier task is valuable only in states where the harder tasks are also completed. As a result, each project is constrained by its *bottleneck*: the task with the lowest safety threshold.

This characterization implies that ownership matters only through its effect on project bottlenecks. Changing ownership changes a project's bottleneck by altering who must be informed for this project to succeed and who can expropriate if informed. Our second main result shows that surplus-maximizing ownership structures maximize the value-weighted sum of bottleneck safety thresholds across projects.

We use this result to derive several implications for ownership patterns. First, when a project requires the information of two parties who are sufficiently tempted to expropriate each other's information, third-party ownership may be optimal. Rather than share information with each other, competitors often outsource the decisions that combine their information to a *knowledge hub*—a third party less tempted to expropriate it (Küpper et al., 2020a).

Second, when the value of one project is sufficiently large relative to others, the optimal ownership structure relaxes the bottleneck in that project. As in our motivating example, when GM's business initially dominated, it was optimal to keep Delphi as a subsidiary to maximize the value of GM's business. As rivals' business grew in value (Helper and Henderson, 2014), the cost of withholding their knowledge rose until the spin-off became optimal.

Extensions. We conclude by considering several variations of the model. First, instead of an omniscient mediator that issues recommendations conditional on the true state, we study a game in which parties contract over a mechanism operated by a neutral uninformed mediator: based on parties' private messages, the mediator enforces an ownership structure and monetary transfers, and issues private action recommendations. We show that the surplus-maximizing mechanism induces the same ownership structure, distribution over states and actions, and total surplus as in our baseline solution concept. Thus, the inability to contract over actions constrains total surplus; the need to elicit private information does not constrain it further.

Second, we show that it is the non-contractibility of expropriation that gives rise to our main trade-offs. When expropriation actions are fully contractible, parties can achieve the first-best surplus (the full value of all projects with no expropriation) regardless of ownership. In contrast, allowing for full contractibility of actions in project tasks or the division of project values does not improve total surplus.

Third, without a mediator, parties can still deter expropriation through cheap talk but cannot coordinate information disclosure across tasks. When coordinated disclosure is unnecessary (as in the spin-off application), the optimal ownership structure is unchanged. When coordinated disclosure matters (as in the knowledge-hub application), the inability to coordinate may shift optimal ownership, suggesting a role for neutral intermediaries — such as consulting or law firms — in raising total surplus.

Fourth, bundling control rights into a single asset implies that the owner must be informed about another party's state for a project to succeed — information the owner can then expropriate — so information is optimally withheld. We show that if control rights can be sufficiently unbundled across multiple assets, the first-best surplus is achievable.

Finally, our main analysis assumes prohibitive expropriation costs, so our optimal decision rule deters expropriation entirely. When costs are not prohibitive, it can be optimal to tolerate some expropriation to enable greater disclosure.

Related Literature. We build on the incomplete-contracts approach to firm boundaries (Grossman and Hart, 1986; Hart and Moore, 1990; Hart, 1995). Grossman and Hart (1986) define ownership as the allocation of residual rights of control: when specific rights are too costly to enumerate contractually, ownership assigns the rights left unspecified. Much of this literature focuses on the allocation of control. We follow it in this respect but also emphasize the *residual* nature of ownership: it typically transfers *multiple* decision rights as a bundle.

Within the firm-boundaries literature, our paper is closest methodologically to work that studies ownership structures through a mechanism-design approach (Klemperer, Cramton, and

Gibbons, 1987; Matouschek, 2004; Segal and Whinston, 2016; Baliga and Sjöström, 2018). Whereas these papers emphasize how ownership affects information rents when ex post actions are contractible, we study optimal ownership when ex post actions are non-contractible.

To characterize the optimal decision rule, we use modern information-design tools (Rayo and Segal, 2010; Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet, Perego, and Taneva, 2020). This approach to study organizational-design questions has also been taken by Kolotilin and Li (2021), who study relational communication in repeated relationships with voluntary transfers and show that selective pooling can be used to manage obedience constraints. All these papers take control structures as given.

A related literature studies how information structures and mediated recommendations shape cooperation in cartels. Sugaya and Wolitzky (2018) show that coarser observability can make collusion easier to sustain because finer information lets firms tailor profitable deviations to market conditions. Marshall and Marx (2007) and Ortner, Sugaya, and Wolitzky (2024) study mediated arrangements that help parties collude, in part through private recommendations and sometimes through correlated randomization. In our paper, related instruments serve as building blocks for a theory of the firm.

We also connect to the literature on communication and delegation in organizations (Dessein, 2002; Alonso, Dessein, and Matouschek, 2008; Rantakari, 2008), which studies how the allocation of control shapes communication. Because ours is a paper on firm boundaries rather than internal organization, we emphasize two features of the environment that delegation models typically abstract from: parties have access to rich contracting instruments (i.e., they can contract jointly on decision rights, information-sharing protocols, and transfers), and the relevant choice is ownership, which allocates a bundle of inextricable control rights.

Finally, our paper relates to the literature on selling information. A central theme in this work is that an informed party faces appropriation hazards when sharing knowledge, and various mechanisms can mitigate them: leveraging competition and credible threats of wider disclosure (Anton and Yao, 1994, 2002), or gradual revelation that relaxes enforcement or informational constraints (Hörner and Skrzypacz, 2016; Garicano and Rayo, 2017; Fudenberg and Rayo, 2019). In our setting, ownership provides an alternative mechanism: it can eliminate appropriation hazards for the owner, but at the cost of creating new ones for others.

2 Model

There are N parties, $i \in \mathcal{N} = \{1, \dots, N\}$, who collaborate on a collection of projects. Each project succeeds only if a set of required actions is correctly tailored to private information held by the parties. Sharing information improves cooperation but creates a risk: parties may use what they learn to expropriate others' knowledge for their private benefit.

Information. The state is $\theta = (\theta_1, \ldots, \theta_N)$, where $\theta_i \in \{-1, 1\}$ is privately observed by party i. The components of θ are independent, and each realization is equally likely, $p(\theta) = \left(\frac{1}{2}\right)^N$.

Actions and Asset Ownership. Each party $i \in \mathcal{N}$ chooses a vector of actions. We distinguish actions by whether they are (i) *cooperative* or *expropriation*, and (ii) *alienable* or *inalienable*. Cooperative actions are productive: when correctly matched to the relevant component of the state, they contribute to project success. Expropriation actions are privately beneficial but socially destructive: matching them to others' information generates a private gain for the acting party while imposing a cost on others. Alienable actions require access to a productive asset, whereas inalienable actions reflect specific human capital and require no such asset.

There is a single productive asset owned by party $g \in \mathcal{N}$. We model control over alienable actions in the following way. Each party $i \in \mathcal{N}$ selects a plan for *alienable cooperative actions*, denoted $x_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and for *alienable expropriation actions*, denoted $z_{ij} \in \{-1, 0, 1\}$ for all $j \neq i$. While every party specifies a plan, only the choices of the asset owner g can be payoff-relevant.

Each party $i \in \mathcal{N}$ also chooses *inalienable cooperation actions*, denoted by $\bar{x}_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and *inalienable expropriation actions*, denoted by $\bar{z}_{ij} \in \{-1, 0, 1\}$ for $j \neq i$.

For each party $i \in \mathcal{N}$, let $a_i = ((x_{ij})_{j \in \mathcal{N}}, (\bar{x}_{ij})_{j \in \mathcal{N}}, (z_{ij})_{j \neq i}, (\bar{z}_{ij})_{j \neq i})$ denote party i's *action plan*. Let $\mathcal{A}_i$ denote the set of possible action plans available to party i and write $\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$ and $a = (a_i)_{i \in \mathcal{N}} \in \mathcal{A}$. Equivalently, we may write $a = (x, \bar{x}, z, \bar{z})$.

Some parties can expropriate only if they own the asset, while others can expropriate even if they do not. Let $\mathcal{E} \subseteq \mathcal{N}$ denote the set of parties whose expropriation opportunities are tied to the asset. We define party i's effective expropriation action against party j as ζ_{ij}. For parties who can intrinsically expropriate ($i \notin \mathcal{E}$), $\zeta_{ij} = \bar{z}_{ij}$. For parties requiring the asset ($i \in \mathcal{E}$), $\zeta_{ij} = z_{ij}$ for the asset owner ($g = i$), and $\zeta_{ij} = 0$ for non-owners ($g \neq i$).

Projects. Production is organized into a set of projects $\mathcal{K}$. Each project $k \in \mathcal{K}$ has value $V_k \geqslant 0$ if successfully implemented and value 0 otherwise. Project success requires that a set of cooperative actions be correctly matched to the relevant components of the state.

Formally, each project k has two sets of requirements. First, $\mathcal{R}_k \subseteq \mathcal{N}$ is the set of alienable requirements: for each $j \in \mathcal{R}_k$, the asset owner g must set $x_{gj} = \theta_j$. Second, $\bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of inalienable requirements: for each $(i, j) \in \bar{\mathcal{R}}_k$, party i must choose $\bar{x}_{ij} = \theta_j$. Given asset owner g, define the project-success indicator $w_k \in \{0, 1\}$ by

$$w_k\left(x, \bar{x}, g; \theta\right) = \left(\prod_{j \in \mathcal{R}_k} \mathbf{1}\{x_{gj} = \theta_j\}\right) \cdot \left(\prod_{(i,j) \in \bar{\mathcal{R}}_k} \mathbf{1}\{\bar{x}_{ij} = \theta_j\}\right).$$

Project k succeeds if and only if all of its alienable and inalienable requirements are met.

Payoffs. Party i's payoff is given by $u_i(a, g; \theta) + t_i$, where t_i is a monetary transfer and

$$u_i(a, g; \theta) = u_i^c(x, \bar{x}, g; \theta) + u_i^e(z, \bar{z}, g; \theta).$$

The term u_i^c represents the *cooperative payoff* from project success, and u_i^e represents the *expropriation payoff.* Transfers satisfy the budget-balance condition, $\sum_{i \in \mathcal{N}} t_i = 0$. We next describe the cooperative and expropriation payoff terms.

The cooperative payoff depends only on which projects succeed. We assume party i receives an exogenous share $\alpha_{ik} \geqslant 0$ of project k's value, with $\sum_{i \in \mathcal{N}} \alpha_{ik} = 1$. Its cooperative payoff is

$$u_i^c(x, \bar{x}, g; \theta) = \sum_{k \in \mathcal{K}} \alpha_{ik} V_k w_k(x, \bar{x}, g; \theta).$$

Thus, cooperative payoffs depend on actions only through project success and do not depend directly on asset ownership beyond its effect on which actions are payoff-relevant.

Expropriation payoffs depend on the effective expropriation actions ζ_{ij}. If party i attempts to expropriate party j's knowledge, and the attempt is successful (i.e., $\zeta_{ij} = \theta_j$), then party i obtains a private benefit $b_{ij} > 0$, while party j incurs a loss $\lambda_{ij} > 0$. If the attempt is unsuccessful (i.e., $\zeta_{ij} = -\theta_j$), party i incurs a cost $c_{ij} > 0$. If $\zeta_{ij} = 0$, both parties receive 0. Party i's expropriation payoff is therefore

$$u_i^e(z, \bar{z}, g; \theta) = \sum_{j \neq i} \left[b_{ij} \mathbf{1}\{\zeta_{ij} = \theta_j\} - c_{ij} \mathbf{1}\{\zeta_{ij} = -\theta_j\} \right] - \sum_{j \neq i} \lambda_{ji} \mathbf{1}\{\zeta_{ji} = \theta_i\},$$

where $\zeta_{ij} = \zeta_{ij}(z_{ij}, \bar{z}_{ij}; g)$ is the effective expropriation action.

We define total surplus as $TS(a, g; \theta) := \sum_{i \in \mathcal{N}} u_i(a, g; \theta)$. Because transfers are budget-balanced, they do not directly enter this expression.

We impose the following two assumptions on expropriation payoffs:

Assumption 1. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $\lambda_{ij} - b_{ij} > V_{\text{total}} := \sum_{k \in \mathcal{K}} V_k$.*

Assumption 2. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $b_{ij} < c_{ij}$.*

The first assumption ensures that parties jointly prefer to avoid expropriation. The second assumption ensures expropriation is risky and thus attractive only when a party has sufficiently precise information about the target's state.

Non-Contractibility of Actions. Following Arrow (1962), we assume that once knowledge is shared, its subsequent use is difficult to restrict contractually. We capture this friction by assuming that actions are *non-contractible.*

Decision Rule. Following the information-design-in-games approach of Bergemann and Morris (2019), we consider an *omniscient* mediator who decides how information is disclosed to the parties. A *decision rule* is a mapping $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$ for each $\theta \in \Theta$, which assigns to each realized state θ a distribution over action profiles $a = (a_i)_{i\in\mathcal{N}}$.

Timing. Fix an ownership structure $g \in \mathcal{N}$. The timing is as follows: (i) the mediator commits to a decision rule σ; (ii) the state θ is realized; (iii) each party i observes θ_i; (iv) the mediator draws an action profile a according to $\sigma(\cdot \mid \theta)$ and privately recommends a_i to each party i; and (v) the parties choose their actions.

Obedience. Recommendations from the mediator are not enforceable. We restrict attention to *obedient* decision rules, under which no party wishes to deviate from the recommended action after observing its private recommendation and private state. We denote the set of obedient decision rules by Σ.

Solution Concept. For a fixed ownership structure $g \in \mathcal{N}$, we characterize an *optimal decision rule* that maximizes total surplus subject to obedience. Let

$$TS(g) := \max_{\sigma\in\Sigma} \mathbb{E}\left[TS(a, g; \theta)\right] \tag{2.1}$$

be the maximum attainable total surplus, where the expectation is taken with respect to the joint distribution $\psi(a, \theta) := p(\theta)\sigma(a \mid \theta)$. An *optimal ownership structure* is any g^* that solves $\max_{g\in\mathcal{N}} TS(g)$.

Discussion of Assumptions. In Section 5.1, we show that the outcome identified by our solution concept can be supported as an equilibrium outcome in a broader contracting game.

The model also relies on several assumptions that we relax in Section 5. We discuss how the results change under the contractibility of actions and project shares α_{ik} (Section 5.2), unmediated cheap-talk communication (Section 5.3), separable control rights across multiple assets (Section 5.4), and arbitrary expropriation costs (Section 5.5).

3 Analysis

We solve the model in two steps. First, we characterize an optimal decision rule for a fixed ownership structure. We show that it is optimal to ensure that no party with the means to expropriate receives enough information to profitably engage in expropriation. Second, we characterize an optimal ownership structure. We show that ownership determines both who must be informed and who has the means to expropriate. All proofs are in the Appendix.

3.1 Optimal Information Disclosure

Fix an ownership structure $g \in \mathcal{N}$ and a decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$. The mediator draws an action profile $a = (a_1, \ldots, a_N)$ from $\sigma(\cdot \mid \theta)$ and privately recommends a_i to each party i. Party i observes its private state θ_i and the recommendations in a_i. The decision rule is obedient ($\sigma \in \Sigma$) if all parties prefer to follow their recommended actions:

$$\sum_{\theta_{-i} \in \Theta_{-i}} \sum_{a_{-i} \in \mathcal{A}_{-i}} p(\theta_{-i} \mid \theta_i)\sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i})\Big[u_i\big((a_i, a_{-i}), g; \theta_i, \theta_{-i}\big) - u_i\big((a_i', a_{-i}), g; \theta_i, \theta_{-i}\big)\Big] \geqslant 0$$

$$\text{for all } i \in \mathcal{N},\ \theta_i \in \Theta_i,\ a_i \in \mathcal{A}_i,\ a_i' \in \mathcal{A}_i. \tag{3.1}$$

The set of inequalities in (3.1) defines obedience constraints in cooperative and expropriation actions. As we show below, the main results are driven by obedience in expropriation actions, which can be characterized using posterior beliefs about states. For $j \neq i$, let

$$\tau_{ij}(\theta_i, a_i) := \mathbb{P}_\psi\big(\theta_j = 1 \,|\, \theta_i, a_i\big)$$

denote the posterior belief of party i that $\theta_j = 1$ after observing (θ_i, a_i). This posterior is defined under the joint distribution $\psi(\theta, a) = p(\theta)\sigma(a|\theta)$. When the dependence on (θ_i, a_i) is clear from context, we write τ_{ij} for $\tau_{ij}(\theta_i, a_i)$.

Suppose party i can engage in expropriation. Then it chooses $\zeta_{ij} = 1$ if $\tau_{ij}b_{ij} - (1-\tau_{ij})c_{ij} > 0$, chooses $\zeta_{ij} = -1$ if $(1-\tau_{ij})b_{ij} - \tau_{ij}c_{ij} > 0$, and chooses $\zeta_{ij} = 0$ otherwise. Therefore, if party i receives a recommendation not to expropriate ($\zeta_{ij} = 0$), its posterior belief must satisfy

$$\max\{\tau_{ij}, 1 - \tau_{ij}\} \leqslant \tau_{ij}^\star := \frac{c_{ij}}{b_{ij} + c_{ij}},$$

where we assume that parties do not expropriate when indifferent. Assumption 2, that is, $c_{ij} > b_{ij}$ for $i \neq j$, implies that $\tau_{ij}^\star > 1/2$. We say that party i is more tempted to expropriate j's information than party ℓ if $\tau_{ij}^\star < \tau_{\ell j}^\star$.

Before stating the general results on the structure of an optimal decision rule, it is useful to build intuition through two examples. The first example shows that it is optimal to disclose information to a party capable of expropriation only up to the point at which it is indifferent between expropriation and no expropriation. The second example shows that it is optimal to correlate recommendations.

Example 1: Optimal Decision Rule For One Party. Suppose there are two parties, $\mathcal{N} = \{1, 2\}$, and one project, $\mathcal{K} = \{1\}$. The project does not require an asset, $\mathcal{R}_1 = \varnothing$, but requires party 1 to match the state of party 2 via its inalienable action, $\bar{\mathcal{R}}_1 = \{(1, 2)\}$. That

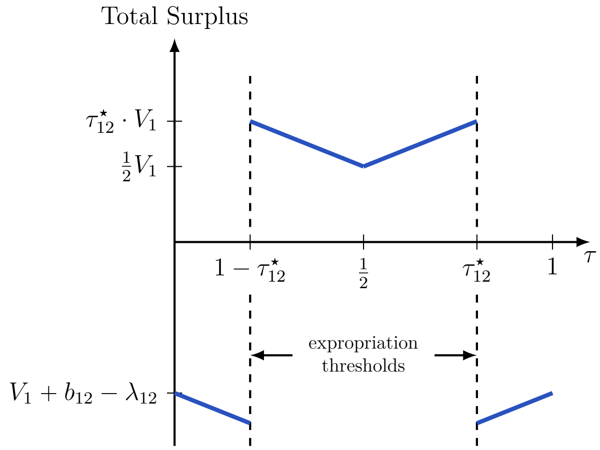

Figure 1: Total Surplus As a Function of Beliefs (Example 1)

This figure illustrates the relationship between total expected surplus and party 1's belief that $\theta_2 = 1$ for Example 1.

is, the project succeeds if and only if $\bar{x}_{12} = \theta_2$. Parties do not require the asset to engage in expropriation, $\mathcal{E} = \varnothing$. Suppose both parties receive half of the project's value, $\alpha_{11} = \alpha_{21} = 1/2$. In this example, the asset ownership is irrelevant.

The optimal decision rule can be found using the standard Bayesian Persuasion argument (Kamenica and Gentzkow, 2011). Denote by τ the posterior belief of party 1 that $\theta_2 = 1$ after receiving the recommendation from the mediator. Figure 1 illustrates the expected total surplus as a function τ. It is optimal for party 1 to select $\bar{x}_{12} = 1$ if $\tau > 1/2$ and $\bar{x}_{12} = -1$ if $\tau < 1/2$. If party 1's belief about θ_2 is extreme, it also optimally engages in expropriation. Specifically, if $\tau > \tau^\star_{12}$, party 1 selects $\zeta_{12} = 1$, and $\tau < 1 - \tau^\star_{12}$, party 1 selects $\zeta_{12} = -1$.

Under expropriation, the total expected surplus is negative. Thus, the optimal decision rule is to randomize between posteriors $\tau^\star_{12}$ and $1 - \tau^\star_{12}$. One way to construct such a belief distribution, which will be useful in the general construction, is as follows. Suppose there is a random variable $\xi \sim \text{Unif}[0, 1]$ independent of θ. The recommendations to party 1 are

$$x_{12} = \bar{x}_{12} = \begin{cases} \theta_2 & \text{if } \xi \leqslant \tau^\star_{12}, \\ -\theta_2 & \text{if } \xi > \tau^\star_{12}, \end{cases} \quad \text{and} \quad z_{12} = \bar{z}_{12} = 0.$$

Example 2: Nested Decision Rule For Multiple Parties. Suppose there are three parties, $\mathcal{N} = \{1, 2, 3\}$, and two projects, $\mathcal{K} = \{1, 2\}$. Neither project has alienable requirements, so $\mathcal{R}_1 = \mathcal{R}_2 = \varnothing$. Project 1 requires two inalienable actions from parties 1 and 2 to match the state of party 3, so $\bar{\mathcal{R}}_1 = \{(1, 3), (2, 3)\}$. Project 2 requires only an inalienable action from party 2 to match the state of party 3, so $\bar{\mathcal{R}}_2 = \{(2, 3)\}$. All parties possess inalienable means

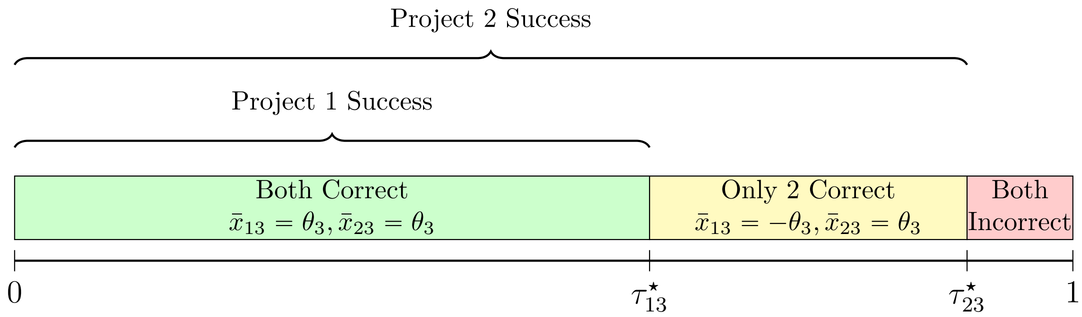

Figure 2: Nested Decision Rule For Example 2

The figure illustrates the realization of recommendations based on the random variable $\xi \sim \text{Unif}[0,1]$. Party 1 receives the correct recommendation only when $\xi \leqslant \tau_{13}^{\star}$. Party 2 receives the correct recommendation when $\xi \leqslant \tau_{23}^{\star}$. This decision rule ensures that Party 2 receives the correct recommendation whenever Party 1 receives the correct recommendation.

of expropriation ($\mathcal{E} = \varnothing$). Finally, suppose party 1 is more tempted to expropriate party 3's information than party 2, so $\tau_{13}^{\star} < \tau_{23}^{\star}$.

To deter expropriation, the information conveyed by party i's recommendation for the action $\bar{x}_{i3}$ must be such that $\max\{\tau_{i3}, 1 - \tau_{i3}\} \leqslant \tau_{i3}^{\star}$. Following Example 1, a natural decision rule is to recommend the correct match to party i with probability $\tau_{i3}^{\star}$ and the incorrect match with probability $1 - \tau_{i3}^{\star}$. However, with two parties required for Project 1, the question is whether these recommendations should be generated independently.

If parties 1 and 2 receive correct recommendations independently, then Project 1 succeeds with probability $\tau_{13}^{\star} \cdot \tau_{23}^{\star}$, while Project 2 succeeds with probability $\tau_{23}^{\star}$. Hence, the expected total surplus is $V_1 \cdot \tau_{13}^{\star} \cdot \tau_{23}^{\star} + V_2 \cdot \tau_{23}^{\star}$.

The mediator can do strictly better by *positively correlating* recommendations. Let $\xi \sim \text{Unif}[0,1]$ be a *common* randomization device. Consider the nested rule in which, for each $i \in \{1,2\}$, the mediator privately recommends $\bar{x}_{i3} = \theta_3$ if $\xi \leqslant \tau_{i3}^{\star}$ and $\bar{x}_{i3} = -\theta_3$ otherwise. This rule is illustrated in Figure 2: Project 1 succeeds with probability $\min\{\tau_{13}^{\star}, \tau_{23}^{\star}\} = \tau_{13}^{\star}$, while Project 2 succeeds with probability $\tau_{23}^{\star}$. The expected total surplus is $V_1 \cdot \tau_{13}^{\star} + V_2 \cdot \tau_{23}^{\star}$, which strictly exceeds the independent benchmark. This decision rule ensures that whenever party 1 is recommended the correct action, party 2 is also recommended the correct action.

Proposition 1 below shows that such nested decision rules — implemented via a common randomization device — are optimal in general.

General Results. Fix an ownership structure $g \in \mathcal{N}$. Let $\text{Arm}(g)$ denote the set of "armed" parties, that is, parties who can engage in expropriation under ownership structure g:

$$\text{Arm}(g) := \{i \in \mathcal{N} : i = g \text{ or } i \notin \mathcal{E}\}.$$

This set consists of the owner and all parties with inalienable means to expropriate ($i \notin \mathcal{E}$).

For each party i and state θ_j with $j \neq i$, we define a restriction on belief τ_{ij} so that party i does not expropriate j's information. If $i \notin \mathrm{Arm}(g)$, then it cannot expropriate for any $\tau_{ij} \in [0,1]$. If $i \in \mathrm{Arm}(g)$, then party i can expropriate but prefers not to do so if $\max\{\tau_{ij}, 1-\tau_{ij}\} \leqslant \tau^{\star}_{ij}$. Accordingly, we define a *safety threshold* as

$$\tau^{\mathrm{safe}}_{ij}(g) := \begin{cases} \tau^{\star}_{ij} & \text{if } i \in \mathrm{Arm}(g) \\ 1 & \text{if } i \notin \mathrm{Arm}(g) \end{cases}.$$

Party i does not expropriate j's information if $\tau_{ij} \in [1-\tau^{\mathrm{safe}}_{ij}(g), \tau^{\mathrm{safe}}_{ij}(g)]$. For notational convenience, we set $\tau^{\mathrm{safe}}_{ii}(g) \equiv 1$ for all $i \in \mathcal{N}$ and $g \in \mathcal{N}$.

For a project $k \in \mathcal{K}$, it is useful to represent the project's requirements as *tasks* of the form "party i must match party j's state" that must be completed. Let

$$\mathcal{Q}_k(g) := \{(g,j) : j \in \mathcal{R}_k\} \ \cup \ \bar{\mathcal{R}}_k \ \subseteq \ \mathcal{N} \times \mathcal{N}$$

denote the set of such tasks that must be completed for project k to succeed under an ownership structure $g \in \mathcal{N}$.

The next proposition establishes that for each project $k \in \mathcal{K}$ the maximum attainable expected surplus is determined by its *bottleneck*: the task with the lowest safety threshold $\tau^{\mathrm{safe}}_{ij}(g)$ among tasks in $(i,j) \in \mathcal{Q}_k(g)$.

Proposition 1 (Optimal Decision Rule). *Fix an ownership structure $g \in \mathcal{N}$. Under Assumptions 1 and 2, the maximum attainable expected total surplus is*

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau^{\mathrm{safe}}_{ij}(g) \right). \tag{3.2}$$

Moreover, there exists an optimal decision rule with a nested structure: whenever a task (i,j) with lower safety threshold is completed, all tasks with higher safety thresholds are completed as well. The mediator draws a single common random variable $\xi \sim \mathrm{Unif}[0,1]$ and, conditional on (θ, ξ), privately recommends:

(i) No expropriation: each party i is recommended to choose $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$.

(ii) Cooperation: party i is recommended the correct match ($x_{ij} = \bar{x}_{ij} = \theta_j$) if and only if $\xi \leqslant \tau^{\mathrm{safe}}_{ij}(g)$, and is recommended the opposite match otherwise ($x_{ij} = \bar{x}_{ij} = -\theta_j$).

The intuition behind Proposition 1 is as follows. In an optimal decision rule, each task $(i,j) \in \mathcal{Q}_k(g)$ is completed with probability at most its safety threshold $\tau^{\mathrm{safe}}_{ij}(g)$. For a disarmed party, this probability is one, since it cannot expropriate and can therefore be fully informed. For an armed party $i \in \mathrm{Arm}(g)$, the success probability of task (i, j) is capped at $\tau^{\star}_{ij}$: any higher, and party i would receive enough information to profitably engage in expropriation, reducing total surplus. Under O-ring complementarities, a project succeeds only if all of its tasks do, so project k succeeds with probability at most $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g)$. The decision rule in Proposition 1 is obedient and attains this upper bound.

3.2 Optimal Asset Ownership

Proposition 1 characterizes the maximum attainable total surplus $TS(g)$ under each ownership structure $g \in \mathcal{N}$. An optimal ownership structure is any $g^* \in \arg\max_g TS(g)$. To characterize how ownership shapes total surplus, it is useful to write $TS(g)$ in terms of its loss relative to the first-best surplus V_{total}.

Corollary 1. *The maximum attainable surplus under an ownership structure $g \in \mathcal{N}$ can be written as*

$$TS(g) = V_{\text{total}} - \sum_{k\in\mathcal{K}} V_k \cdot \max_{i\neq j} \left\{ \mathbf{1}\{(i,j) \in \mathcal{Q}_k(g)\} \cdot \mathbf{1}\{i \in \mathrm{Arm}(g)\} \cdot \left(1 - \tau^{\star}_{ij}\right) \right\}. \tag{3.3}$$

The first-best surplus V_{total} is attainable if and only if there exists an ownership structure $g \in \mathcal{N}$ such that no armed party is required to match another party's state in any project.

This corollary shows that ownership can affect total surplus through two distinct channels. First, it can affect tasks by changing who is responsible for taking alienable cooperative actions. Second, it can affect the set of armed parties by changing who has the means to expropriate. If ownership affects neither the set of tasks nor the set of armed parties, then it has no effect on total surplus. The next two examples isolate these channels.

Example 3: Ownership and Tasks. Suppose there are two parties, $\mathcal{N} = \{1, 2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties ($\mathcal{R}_1 = \{1, 2\}$); no inalienable actions are required ($\bar{\mathcal{R}}_1 = \varnothing$); and every party has the inalienable means to expropriate ($\mathcal{E} = \varnothing$).

In this example, both parties are armed regardless of ownership, $\mathrm{Arm}(g) = \mathcal{N}$ for any $g \in \{1, 2\}$. Ownership affects only who must match another party's state. The set of tasks under owner g is $\mathcal{Q}_1(g) = \{(g, 1), (g, 2)\}$. Since the owner can always match its own state, the only task requiring disclosure is (g, j), where $j \neq g$. Thus,

$$TS(g) = V_1 - V_1 \max\Big\{ \underbrace{\mathbf{1}\{(g,j) \in \mathcal{Q}_1(g)\}}_{=1} \cdot 1 \cdot (1 - \tau^{\star}_{gj}),\ \underbrace{\mathbf{1}\{(j,g) \in \mathcal{Q}_1(g)\}}_{=0} \cdot 1 \cdot (1 - \tau^{\star}_{jg})\Big\} = \tau^{\star}_{gj} \cdot V_1,$$

so optimal ownership assigns the asset to the party with the lower temptation to expropriate.

Example 4: Ownership and Expropriation. Suppose there are two parties, $\mathcal{N} = \{1, 2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties ($\mathcal{R}_1 = \{1, 2\}$), and, in addition, requires *inalienable* cooperative actions: each party must match the other party's state ($\bar{\mathcal{R}}_1 = \{(1, 2), (2, 1)\}$). Suppose also that both parties require the asset to engage in expropriation ($\mathcal{E} = \{1, 2\}$).

In this example, both parties must match each other's state regardless of ownership: $(i, j) \in \mathcal{Q}_1(g)$ for all $i \neq j$ and $g \in \mathcal{N}$. Ownership therefore does not determine whose information can be withheld without affecting total surplus. Instead, it determines who is armed: $\mathrm{Arm}(g) = \{g\}$. The resulting surplus is

$$TS(g) = V_1 - V_1 \max\left\{1 \cdot \underbrace{\mathbf{1}\{g \in \mathrm{Arm}(g)\}}_{=1} \cdot (1 - \tau^\star_{gj}),\ 1 \cdot \underbrace{\mathbf{1}\{j \in \mathrm{Arm}(g)\}}_{=0} \cdot (1 - \tau^\star_{jg})\right\} = \tau^\star_{gj} \cdot V_1,$$

so an optimal ownership structure is the same as in Example 3. But the reason is different: rather than determining who must receive information for the project to succeed, ownership determines who lacks the means to expropriate.

These examples illustrate how ownership can protect the owner's information, in the sense that tasks involving the owner's state are completed with probability one and without expropriation. In Example 3, the owner can complete the task by matching its own state without disclosing it to others. In Example 4, the owner's information can be safely disclosed to another party because that party lacks the means to expropriate it.

Another implication of Corollary 1 is that if at least one project has two alienable requirements, $|\mathcal{R}_k| \geqslant 2$ for some $k \in \mathcal{K}$, then the first-best surplus is unattainable. In this case, an asset owner is responsible for matching at least one state of *another* party. Since the owner is always armed, it cannot be fully informed about that state, reducing total surplus. This implication highlights how bundling control over multiple decisions into a single asset is a source of inefficiency. We return to this point in Section 5.4.

Ownership and Temptation to Expropriate. Next, we derive how an optimal ownership structure depends on parties' temptation to expropriate. Recall that for an armed party i and target $j \neq i$, the safety threshold is given by $\tau^\star_{ij} = \frac{c_{ij}}{b_{ij} + c_{ij}}$, where c_{ij} is the cost of unsuccessful expropriation and $b_{ij} < c_{ij}$ is the benefit of successful expropriation. Consider a *temptation shock* $h \in [0, 1)$ that increases the benefit of successful expropriation to $b_{ij}(h) = b_{ij} + h \cdot (c_{ij} - b_{ij})$.

Suppose the temptation shock affects only a subset of party pairs. Let $\mathcal{Z} \subseteq \{(i, j) \in \mathcal{N}^2 : i \neq j\}$ denote the set of affected pairs. For $(i, j) \in \mathcal{Z}$, the benefit of successful expropriation becomes $b_{ij}(h)$ with $h > 0$, while for $(i, j) \notin \mathcal{Z}$ it remains $b_{ij}(0) = b_{ij}$.

We say that an ownership structure $g \in \mathcal{N}$ is $\mathcal{Z}$*-insulated* if, for every affected pair $(i, j) \in \mathcal{Z}$ and every project $k \in \mathcal{K}$, either i is not required to match j's state or i is disarmed:

$$\mathbf{1}\{(i, j) \in \mathcal{Q}_k(g)\} \cdot \mathbf{1}\{i \in \mathrm{Arm}(g)\} = 0 \quad \text{for all } (i, j) \in \mathcal{Z} \text{ and } k \in \mathcal{K}.$$

We prove the following result.

Corollary 2. *If a $\mathcal{Z}$-insulated ownership structure is optimal under the temptation shock $\bar{h}$, then it remains optimal for all higher shocks $h > \bar{h}$.*

Higher temptation to expropriate can decrease total surplus only through tasks in which an armed party must match another party's state. An insulated ownership structure is a way to organize tasks to avoid exposure to the temptation shock.

Ownership and Project Value. Next, we show that an increase in the value of project $\ell \in \mathcal{K}$ shifts optimal ownership toward structures that minimize the loss in that project. Define the loss in project k as

$$\mathrm{Loss}_k(g) := \max_{i \neq j} \left\{ \mathbf{1}\{(i, j) \in \mathcal{Q}_k(g)\} \cdot \mathbf{1}\{i \in \mathrm{Arm}(g)\} \cdot (1 - \tau^{\star}_{ij}) \right\},$$

and let $\Gamma_\ell := \arg\min_{g \in \mathcal{N}} \mathrm{Loss}_\ell(g)$ denote the set of ownership structures that minimize the loss in project ℓ.

Corollary 3. *Holding the values of all projects except ℓ fixed, there exists $\bar{V}_\ell$ such that for every $V_\ell > \bar{V}_\ell$, every optimal ownership structure belongs to Γ_ℓ.*

An equivalent formulation of Corollary 3 is that once one project is sufficiently valuable relative to the others, it is optimal to alleviate the bottleneck in that project: if V_ℓ / V_k is sufficiently large for all $k \neq \ell$, $g^* \in \arg\max_g \{\min_{(i,j) \in \mathcal{Q}_\ell(g)} \tau^{\mathrm{safe}}_{ij}(g)\}$.

4 Applications

This section applies the model to three questions about firm boundaries: when competing firms should outsource decisions that depend on their information to a third party, when a supplier should be independent rather than owned by a customer, and how acquisitions and ownership shape knowledge transfer.

4.1 Knowledge Hubs

A growing form of interfirm cooperation is data sharing aimed at improving products and manufacturing processes (Küpper et al., 2020a). One example is predictive maintenance, which uses data from sensors embedded in industrial equipment together with machine-learning algorithms to predict equipment failures before they occur. Predictive maintenance becomes more effective when algorithms are trained on data pooled from multiple firms facing similar operating environments. Yet these firms are often direct competitors and therefore may be tempted to expropriate each other's information. As a result, firms often outsource predictive maintenance services to third parties.[2]

Our model speaks directly to this issue. Consider an extension of Example 3 with three parties, $\mathcal{N} = \{1, 2, 3\}$, and a single project, $\mathcal{K} = \{1\}$. The project succeeds only if the asset owner matches the states of parties 1 and 2, $\mathcal{R}_1 = \{1, 2\}$. No inalienable actions are required, $\bar{\mathcal{R}}_1 = \varnothing$, and all parties possess the inalienable means to expropriate, so $\mathcal{E} = \varnothing$.

We interpret parties 1 and 2 as competitors with high temptations to expropriate each other's information. The third party neither contributes information nor performs any inalienable actions. Its value stems from its lower temptation to expropriate. By Proposition 1, total surplus under $g \in \{1, 2, 3\}$ is

$$TS(g = 1) = \tau^\star_{12} \cdot V_1, \qquad TS(g = 2) = \tau^\star_{21} \cdot V_1, \qquad TS(g = 3) = \min\{\tau^\star_{31}, \tau^\star_{32}\} \cdot V_1.$$

Thus, the third-party ownership is optimal if $\min\{\tau^\star_{31}, \tau^\star_{32}\} \geqslant \max\{\tau^\star_{12}, \tau^\star_{21}\}$. In this case, party 3 acts as a *knowledge hub*, enabling greater information pooling among competing firms.

4.2 Spin-Offs

In this section, we study the benefits and costs of vertical integration from a knowledge-sharing perspective. Our discussion is motivated by two examples. The first is GM's decision to spin off Delphi Automotive, discussed in the introduction. The second is a recent argument that Intel should spin off its foundry business to make it a more attractive supplier to chip-design companies (Fitch, 2025). Historically, chip design and manufacturing were often performed by the same firm. Over time, the industry shifted toward a structure in which design and manufacturing are separated, with production concentrated in a few large foundries that serve multiple chip-design companies. A common explanation is that foundries realize economies of scale by pooling orders from many design firms (Goldberg et al., 2024). But economies of scale explain why firms benefit from a common manufacturer, not why that manufacturer should be independent rather than owned by one of its customers.

We consider two competing firms and one common supplier and ask whether the supplier

[2] "To implement both applications, the most feasible option seems to be for manufacturers to work with a third party (such as a machine supplier or service provider) to combine, clean and analyse data. This is because, by necessity, the data-sharing arrangements involve companies in the same industry that might be in direct competition with one another" (Küpper et al., 2020b, p. 7).

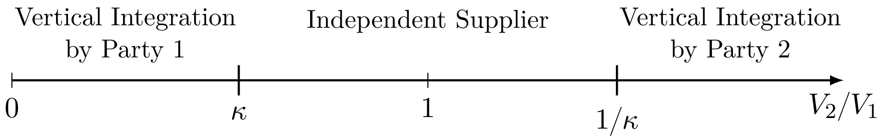

Figure 3: Optimal Ownership When Both Projects Require the Asset

The optimal owner varies with the relative market size V_2/V_1. For this figure, we assume that $\kappa < 1$.

should be independent or owned by one of the competing firms. Let $\mathcal{N} = \{1, 2, 3\}$ and $\mathcal{K} = \{1, 2\}$, with parties 1 and 2 the downstream competitors and party 3 the common upstream supplier. Project 1 covers party 1's operations and project 2 party 2's. Each project has an inalienable requirement that the supplier match the downstream firm's state: $\bar{\mathcal{R}}_1 = \{(3, 1)\}$ and $\bar{\mathcal{R}}_2 = \{(3, 2)\}$. Each also has an alienable requirement that the asset owner match that state: $\mathcal{R}_1 = \{1\}$ and $\mathcal{R}_2 = \{2\}$.

If a downstream party owns the asset ($g \in \{1, 2\}$), we refer to the supplier as that party's subsidiary, that is, they are vertically integrated. If the supplier owns the asset ($g = 3$), we refer to it as an independent upstream firm.

If one of the downstream firms owns the asset ($g \in \{1, 2\}$), the owner's project succeeds with certainty: the owner's state is successfully matched by the owner and by the disarmed supplier. The surplus in the non-owner's project is constrained because the armed owner must match the non-owner's state, so information about that state is optimally partially withheld. If the supplier owns the asset, both projects are constrained by the supplier's temptation to expropriate. The maximum attainable surplus under each ownership structure is

$$TS(g = 1) = V_1 + \tau^\star_{12} V_2, \qquad TS(g = 2) = \tau^\star_{21} V_1 + V_2, \qquad TS(g = 3) = \tau^\star_{31} V_1 + \tau^\star_{32} V_2.$$

To simplify the derivation of an optimal ownership structure, suppose that $\tau^\star_D := \tau^\star_{12} = \tau^\star_{21}$ and $\tau^\star_U := \tau^\star_{31} = \tau^\star_{32}$. Suppose also that the supplier is less tempted to expropriate than are the downstream competitors, that is, $\tau^\star_U > \tau^\star_D$. Define the threshold

$$\kappa := \frac{1 - \tau^\star_U}{\tau^\star_U - \tau^\star_D}$$

and assume that $\kappa < 1$. Then vertical integration by party 1 ($g = 1$) is optimal when its project is sufficiently more valuable than party 2's project, $V_2/V_1 < \kappa$. Symmetrically, if $V_2/V_1 > 1/\kappa$, vertical integration by party 2 ($g = 2$) is optimal. For intermediate values, $V_2/V_1 \in [\kappa, 1/\kappa]$, an independent supplier ($g = 3$) is optimal.[3] Figure 3 summarizes the resulting pattern.

[3] If $\kappa > 1$, the optimal ownership is $g^* = 1$ if $V_2/V_1 \leqslant 1$ and $g^* = 2$ if $V_2/V_1 > 1$.

GM's shift from vertical integration to spinning off its parts supplier can be explained by the growing value of other automakers' business relative to GM's. When GM dominated, integration was optimal.[4] As rivals' business became more valuable (Helper and Henderson, 2014), the loss from restricting their information exceeded the benefit of full knowledge sharing between GM and the supplier, making the spin-off optimal. This logic depends on all parties being tempted to expropriate. If one party had no temptation, it would be optimal to assign the asset to that party regardless of the relative value of GM's and rivals' businesses.

The same trade-off may underlie the case for Intel to spin off its foundry business. A spin-off would likely encourage greater knowledge sharing from rival design firms, which may be reluctant to disclose proprietary information to a foundry owned by a competitor. At the same time, it would weaken Intel's incentives to share knowledge with the now-independent foundry.

4.3 Acquisitions, Ownership, and Knowledge Transfer

Recent empirical work on firm boundaries documents a regularity: after acquisitions, acquired firms often come to resemble their acquirers along a range of organizational and operational dimensions (Atalay, Hortaçsu, and Syverson, 2014; Braguinsky et al., 2015; Eliason et al., 2020; La Forgia and Bodner, 2024; Demirer and Karaduman, 2025). These findings are typically interpreted as evidence that integration facilitates knowledge transfer.[5]

Our model provides an explanation for this pattern. If an acquisition removes the acquired party's ability to expropriate, as in Example 4 of Section 3.2, the acquirer can safely transfer more knowledge to the acquired party. Apple's operations in China illustrate this mechanism. Although Apple outsourced production to contract manufacturers, it retained ownership of key production equipment. Ownership of this equipment limited suppliers' ability to use Apple's information outside the relationship and thereby reduced the risk of expropriation. As a result, Apple could safely transfer proprietary engineering techniques to its suppliers (McGee, 2025).

The model however highlights that removing the ability to expropriate is not the only way ownership can protect the owner's information. Example 3 in Section 3.2 illustrates how ownership can allow the owner to apply its information directly without disclosure to another party. For example, Apple chose to design the iPad chip in-house rather than partner with Intel because it "just didn't want to teach them everything, which they could go and sell to our competitors" (Steve Jobs, quoted in Isaacson, 2011, p. 493).

Our model also emphasizes that ownership has costs: non-owners may not be willing to

[4] Delphi Automotive consolidated several parts suppliers owned by GM; we frame the discussion around a single supplier to simplify exposition.

[5] "This analysis contributes to the growing body of evidence suggesting that acquisitions serve as a mechanism for transferring within-organization knowledge to newly acquired assets" (Demirer and Karaduman, 2025, p. 4).

fully disclose their information to the owner. For example, China's quid pro quo policy required multinationals to form joint ventures with local partners to access the Chinese market. This structure gave the local partner control over some manufacturing decisions, forcing the multinational to disclose production-relevant information. One implication of this policy was that it could discourage automakers from bringing their most advanced technology to China due to concerns about expropriation (Bai et al., 2025).

5 Discussion

This section clarifies the mechanisms driving our main results by exploring several variations of the baseline model. We establish that the outcome of our information-design approach can implemented in a broader contracting game (Section 5.1), isolate the specific contracting frictions that make ownership matter (Section 5.2), and relax the assumptions of mediated communication (Section 5.3), bundled control rights (Section 5.4), and prohibitive expropriation costs (Section 5.5). All proofs are in the Appendix.

5.1 Implementation Through a Mediated Contract

In the baseline model, we assume that an omniscient mediator can condition recommendations on the true state, and ownership is chosen to maximize total surplus. Suppose instead that the mediator does not observe the state directly, but parties can use the mediator to exchange information and enforce contracts on ownership and transfers.

Direct Mediated Mechanism. A *direct mediated mechanism* consists of an allocation rule $\varphi : \Theta \to \Delta(\mathcal{A} \times \mathcal{N})$ and a transfer rule $t : \mathcal{A} \times \mathcal{N} \times \Theta \to \mathcal{T}$, where $\Theta := \{-1, 1\}^N$ is the state space and $\mathcal{T} = \{t \in \mathbb{R}^N : \sum_i t_i = 0\}$ is the set of budget-balanced transfer vectors. An allocation rule maps each state to a distribution over action recommendations and ownership structures. A transfer rule maps action recommendations, ownership structures, and states into a vector of transfers. In what follows, our restriction to direct mediated mechanisms is without loss of generality (Forges, 1986).

Communication Game. Given a direct mediated mechanism (φ, t), consider the following communication game. (i) The state θ is realized, and each party i privately observes θ_i. (ii) Each party i simultaneously submits a confidential report $\hat{\theta}_i$ to a neutral mediator. The mediator observes $\hat{\theta} := (\hat{\theta}_1, \ldots, \hat{\theta}_N)$ and then enforces an ownership structure $g \in \mathcal{N}$ and sends private action recommendations according to $\varphi(\hat{\theta})$. The mediator also enforces the transfer vector according to $t(a, g, \hat{\theta})$. (iii) Each party privately observes its action recommendation

$a_i \in \mathcal{A}_i$, the enforced ownership structure $g \in \mathcal{N}$, and the vector of transfers $t \in \mathcal{T}$. (iv) Parties simultaneously choose their non-contractible actions.

An outcome is a joint distribution over actions, ownership, transfers, and states. An outcome is feasible if there exists a direct mediated mechanism and a Bayesian Nash equilibrium of the corresponding communication game that induces it. To characterize the set of all feasible outcomes, it suffices to characterize the set of incentive-compatible direct mediated mechanisms (Forges, 1986). The incentive-compatibility constraints consist of (i) truth-telling constraints, that is, parties cannot unilaterally gain by misreporting their state, (ii) obedience constraints, that is, parties cannot unilaterally gain by deviating from their recommended actions, and (iii) double-deviation constraints, that is, parties cannot unilaterally gain by misreporting their states and deviating from the recommended actions.

We characterize an *optimal* mechanism, that is, an incentive-compatible direct mediated mechanism that maximizes total surplus. Consider the following allocation rule:

$$\varphi^*(a, g \mid \theta) = \begin{cases} \sigma^*(a \mid \theta), & \text{if } g = g^*, \\ 0, & \text{if } g \neq g^*, \end{cases} \tag{5.1}$$

where g^* is an optimal ownership structure that maximizes $TS(g)$ and $\sigma^*(a \mid \theta)$ is the optimal decision rule specified under g^* in Proposition 1.

Proposition 2. *Any mechanism that consists of allocation rule φ^* and any constant budget-balanced transfers is optimal and yields surplus $TS(g^*)$.*

The proof of Proposition 2 proceeds in four steps. First, we consider a relaxed problem dropping truth-telling and double-deviation constraints. Second, in this relaxed problem, we show that deterministic ownership is without loss of optimality. Third, we show that this relaxed problem coincides with our baseline problem of characterizing an optimal decision rule and ownership structure. The allocation rule in (5.1) solves this relaxed problem and the resulting surplus is an upper bound on the surplus in the full problem. Finally, we show the mechanism consisting of this allocation rule and constant budget-balanced transfers is incentive compatible and therefore optimal.

The intuition behind this result is as follows. The optimal decision rule ensures that no armed party receives enough information to engage in expropriation. Once this condition is satisfied, the remaining incentives are aligned, and truthful reporting can be sustained. Thus, the central friction in our setting is the non-contractibility of actions rather than private information.

Contracting Game. We now embed the communication game in a broader bargaining environment. Fix an initial owner $g_0 \in \mathcal{N}$. A *contract* is a direct mediated mechanism (φ, t), where φ is an allocation rule and t is a budget-balanced transfer rule, as defined above.

The contracting game has two additional stages before the communication game. First, party 1 makes a take-it-or-leave-it offer of a contract (φ, t) to the other parties. Second, each party $i \in \{2, \ldots, N\}$ either accepts or rejects. If all parties accept, the communication game induced by (φ, t) is played. If at least one party rejects, the asset owner remains g_0, transfers are zero, no recommendations are given, parties choose actions and receive disagreement payoffs.

Proposition 5 in Appendix A.3 shows that there exists a Perfect Bayesian Equilibrium of the contracting game in which party 1 offers the contract (φ^*, t^*), where φ^* is given by (5.1) and t^* ensures that the ex ante utilities for parties 2 to N are equal to their disagreement payoffs. All parties accept the contract, and, conditional on acceptance, parties report truthfully and obey their recommendations.

5.2 Contractibility of Actions

In the baseline model, we assume that all actions are non-contractible. If some actions are contractible, then whenever the mediator recommends those actions they are enforced, which relaxes the obedience constraints. Thus, contractibility weakly increases the maximum attainable total surplus. This section outlines the intuition behind the results; formal results are given Appendix A.4.

Expropriation. If *expropriation* actions $(z, \bar{z})$ are contractible, then the mediator can directly enforce $z_{ij} = \bar{z}_{ij} = 0$ before any information is disclosed. Once expropriation is ruled out, the mediator can fully reveal information and achieve the first best under *any* ownership structure.

We also consider partial contractibility of expropriation. Fix an ownership structure $g \in \mathcal{N}$. For simplicity, suppose all parties require the asset to engage in expropriation, $\mathcal{E} = \mathcal{N}$. If party i is disarmed, then the effective expropriation action is $\zeta_{ij} = 0$, as in the baseline model. If i is armed, then suppose it can take two types of alienable expropriation actions actions, z^1_{ij} and z^2_{ij}. The action z^1_{ij} is non-contractible, while z^2_{ij} is contractible. The effective expropriation action is

$$\zeta_{ij} = \begin{cases} z^1_{ij} & \text{with probability } q^c, \\ z^2_{ij} & \text{with probability } 1 - q^c. \end{cases}$$

Thus, $q^c = 1$ corresponds to the baseline case of non-contractible expropriation actions, while $q^c = 0$ corresponds to full contractibility of expropriation.

Since z^2_{ij} is contractible, it is optimal to specify $z^2_{ij} = 0$. The remaining non-contractible action z^1_{ij} affects payoffs only with probability q^c. For any $q^c > 0$, suppose the analogue of Assumption 1 holds: $q^c \cdot (\lambda_{ij} - b_{ij}) > V_{\text{total}}$ for all $i \neq j$. Then, an optimal deters expropriation and discloses information only up to the same safety thresholds as in the baseline model. If

$TS^q(g)$ denotes the maximum attainable total surplus in this variant, then $TS^q(g) = TS(g)$ for every ownership structure g. When the analogue of Assumption 1 does not hold, it may instead be optimal to tolerate some expropriation in order to enable more information sharing; Section 5.5 studies optimal decision rules in that case.

Cooperation. If either cooperative actions or project shares, or both are contractible, while expropriation remains non-contractible, then the maximum attainable surplus is still $\max_g TS(g)$. To see why this is the case, note that there are two sets of obedience constraints: those involving expropriation actions and those involving cooperative actions. Consider a relaxed problem in which we drop the obedience constraints in cooperative actions. In this problem, the no-expropriation constraints still bind for armed players, resulting in the same total surplus as in our baseline model. When cooperative actions are contractible, the maximum attainable surplus corresponds to the value of this relaxed problem. When only project shares are contractible, the maximum attainable surplus is weakly lower than the value of this problem, and this upper bound is attainable.

5.3 Non-Contractible Communication: Unmediated Cheap Talk

In the baseline setting, the mediator plays two distinct roles: it garbles information to deter expropriation and coordinates disclosure by correlating recommendations across parties. In this subsection, we examine the role of the omniscient mediator by contrasting our baseline model with one in which parties can only engage in *unmediated cheap talk*.

We show that, in an extension with richer state and action spaces, unmediated cheap talk can replicate the mediator's information-garbling role without commitment to randomization. Thus, in environments that require only garbling, such as the spin-off application, the total surplus and an optimal ownership structure are unchanged. However, the mediator's coordination role, which is important in the knowledge-hub application, generally cannot be replicated by unmediated cheap talk.

In our baseline model, for a task (i, j) and an armed party i, the optimal decision rule recommends the correct cooperative action with probability $\tau_{ij}^\star$ and the incorrect action with probability $1 - \tau_{ij}^\star$. With binary states and actions, this garbling requires randomization and hence commitment (Kamenica and Lin, 2025). A pure cheap-talk strategy can only pool or separate the two states. To obtain an intermediate posterior $\tau_{ij}^\star \in (1/2, 1)$, party j would have to randomize after observing its state. Without commitment, this randomization is not credible in our setting unless party i expropriates with positive probability.

Rich State Space: Continuum Analogue. The failure of unmediated cheap talk to achieve optimal garbling is an artifact of the binary state space. Suppose each party i privately observes an independent state θ_i uniformly distributed on a unit circle $\mathbb{T} = [0, 1)$. Cooperative actions x_{ij} and $\bar{x}_{ij}$ are chosen from $\mathbb{T}$. Expropriation actions z_{ij} and $\bar{z}_{ij}$ are chosen from $\mathbb{T} \cup \{\varnothing\}$, where $\varnothing$ denotes the safe action of not attempting expropriation.

An action matches a state if it falls within a target-specific circular tolerance $\delta_j > 0$ of the true state, using the circular distance $d(x, y) := \min\{|x - y|,\ 1 - |x - y|\}$. Specifically, for an owner $g \in \mathcal{N}$, an alienable task $j \in \mathcal{R}_k$ is completed if $d(x_{gj}, \theta_j) \leqslant \delta_j$, and an inalienable task $(i, j) \in \bar{\mathcal{R}}_k$ is completed if $d(\bar{x}_{ij}, \theta_j) \leqslant \delta_j$. As in the baseline model, party i's effective expropriation action is ζ_{ij}. Expropriation succeeds if $\zeta_{ij} \in \mathbb{T}$ and $d(\zeta_{ij}, \theta_j) \leqslant \delta_j$, yielding the same payoffs for successful and unsuccessful attempts as in the baseline model.

We impose the following continuous analogue of Assumption 2:

Assumption 2*. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $b_{ij} < c_{ij}$ and $\delta_j \in (0, 1/4]$.*

This assumption ensures that an uninformed party has no incentive to attempt expropriation. Specifically, under the uniform prior, if an uninformed party i guesses the state θ_j, it succeeds with probability $2\delta_j$. If $\delta_j \leqslant \frac{1}{4}$, the probability of successful expropriation does not exceed $\frac{1}{2}$. Given that $b_{ij} < c_{ij}$, an uniformed party will not attempt to expropriate.

We restrict attention to environments in which each sender's state must be communicated to at most one *armed* party. Formally, fix an owner g and suppose that for every target $j \in \mathcal{N}$, there is at most one party $i \in \mathrm{Arm}(g)$ such that $(i, j) \in \bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)$. This assumption covers Examples 3 and 4 and all applications from Section 4. We prove the following result.

Proposition 3 (Garbling with unmediated cheap talk)**.** *Consider the continuum analogue of our environment. Suppose that each target state must be communicated to at most one armed party. Under Assumptions 1 and 2*, there exists a pure-strategy Perfect Bayesian Equilibrium such that every task (i, j) is completed with probability $\tau^{\text{safe}}_{ij}(g)$, and there is no expropriation on the equilibrium path.*

The PBE is constructed via deterministic pooling of states, as illustrated in Figure 4. For each task (i, j) with an armed receiver i, the state space is partitioned into a success region $[0, \tau^\star_{ij}]$ and a failure region $(\tau^\star_{ij}, 1)$, where recall that $\tau^\star_{ij} = \frac{c_{ij}}{b_{ij} + c_{ij}}$. Both regions are divided into M intervals $\{S^m\}_{m=1}^M$ and $\{F^m\}_{m=1}^M$ such that $\bigcup_{m=1}^M S^m = [0, \tau^\star_{ij}]$ and $\bigcup_{m=1}^M F^m = (\tau^\star_{ij}, 1)$, where $M = \lceil \frac{\tau^\star_{ij}}{2\delta_j} \rceil$ is the number of messages j can send to i.

In equilibrium, party j sends message $m \in \{1, 2 \ldots, M\}$ to i if $\theta_j \in S^m \cup F^m$. Party i's posterior belief is that θ_j is uniformly distributed on $S^m \cup F^m$. The lengths of the success and failure intervals are chosen such that

$$\frac{|S^m|}{|S^m| + |F^m|} = \tau^\star_{ij}.$$

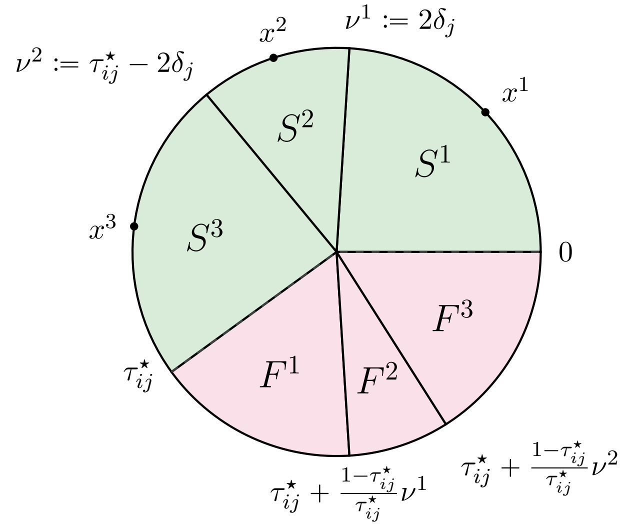

Figure 4: State space partition for the cheap-talk PBE

Party j sends message $m \in \{1, \ldots, M\}$ to armed party i if $\theta_j \in S^m \cup F^m$. This example is constructed under $\delta_j = 0.12$ and $\tau^\star_{ij} = 0.6$, where $M = \lceil \frac{\tau^\star_{ij}}{2\delta_j} \rceil = 3$. The cooperative actions corresponding to message $m \in \{1, 2, 3\}$ are $x^1 = \delta_j$, $x^2 = \tau^\star_{ij}/2$, and $x^3 = \tau^\star_{ij} - \delta_j$.

As Figure 4 shows, the construction of success and failure intervals is such that they are not contiguous, and party i optimally selects an action $x^m \in S^m$ such that the δ-neighborhood of x^m covers the success interval but not the failure interval. Party i is also indifferent between attempting to expropriate and not. In addition, party j has no incentive to deviate: if $\theta_j \in S^m$, then sending m ensures the task (i, j) is completed; if $\theta_j \in F^m$, the task fails regardless of the message sent, making the sender indifferent across all messages.

The Mediator's Problem. The characterization of the maximum attainable surplus in Proposition 1 extends directly to this setting. Specifically, under $g \in \mathcal{N}$, the upper bound on the surplus is $TS(g)$ defined in (3.2), as in our baseline model. An obedient decision rule that attains this surplus is as follows. Define the *safety arc length* as

$$L_{ij}(g) := \frac{2\delta_j}{\tau^{\text{safe}}_{ij}(g)},$$

and let $\xi \sim \text{Unif}[0, 1]$ be a common randomization device. For tasks of the form (i, i), the mediator recommends $x_{ii} = \bar{x}_{ii} = \theta_i$. For tasks of the form (i, j) with $j \neq i$, the mediator recommends

$$x_{ij} = \bar{x}_{ij} = \theta_j + L_{ij}(g)\Big(\frac{1}{2} - \xi\Big) \pmod 1, \qquad z_{ij} = \bar{z}_{ij} = \varnothing.$$

Geometrically, the mediator creates an arc of length $L_{ij}(g)$ centered at the true state θ_j, and then draws a recommendation uniformly at random from this arc. Conditional on receiving this recommendation, the posterior of any party is uniform over an interval of length $L_{ij}(g)$ with the recommended action as its midpoint. Therefore, the maximum probability with which any task is completed or an expropriation attempt succeeds is $2\delta_j/L_{ij}(g) = \tau_{ij}^{\text{safe}}(g)$, which deters expropriation.

In Examples 3 and 4 in Section 3.2 and in the spin-off application in Section 4.2, the PBE constructed in Proposition 3 delivers the same total surplus as the optimal mediated decision rule. Because each project in those environments relies on at most one task with an armed receiver, the optimal decision rule involves only information garbling, which can be replicated through unmediated cheap talk in this rich environment.

The knowledge-hub application in Section 4.1, however, is different. When the supplier acts as the hub ($g = 3$), it must receive information about both θ_1 and θ_2. Proposition 3 ensures that both tasks $(3, 1)$ and $(3, 2)$ can be independently implemented with success probabilities $\tau_{31}^\star$ and $\tau_{32}^\star$. However, because parties can condition their messages only on their own private states, their messages are necessarily independent. This lack of coordination yields a strictly lower expected surplus:

$$TS^{\text{cheap talk}}(g = 3) = \tau_{31}^\star \cdot \tau_{32}^\star \cdot V_1 < \min\{\tau_{31}^\star, \tau_{32}^\star\} \cdot V_1 = TS(g = 3).$$

Thus, unmediated cheap talk cannot replicate mediated coordination of recommendations. Moreover, Proposition 7 in Appendix A.5 shows that, in the knowledge-hub application, such coordination does not require randomization. Instead of drawing $\xi \sim \text{Unif}[0, 1]$, the mediator can use one sender's state to coordinate disclosure from both senders to the hub. The maximum total surplus can therefore be attained by a *deterministic* decision rule.

5.4 Multiple Assets and Separation of Control Rights

In the baseline model, a single asset bundles multiple alienable decision rights. This subsection extends the model to multiple assets and allows control rights to be unbundled.

Assets. Let $\mathcal{M}$ be a finite set of assets. Two assignment functions determine which asset controls each *alienable* decision:

$$\phi^c : \mathcal{N} \to \mathcal{M}, \qquad \phi^e : \mathcal{N} \to \mathcal{M}.$$

Here $\phi^c(j)$ is the asset whose owner controls the alienable *cooperative* action targeting state θ_j, while $\phi^e(j)$ is the asset whose owner controls the alienable *expropriation* action targeting θ_j (for asset-dependent expropriators). The baseline model with one asset corresponds to $|\mathcal{M}| = 1$, so a single party controls *all* alienable actions. In general, control over cooperative actions is bundled when $\phi^c(i) = \phi^c(j)$ for $i \neq j$; control over cooperative and expropriation actions is bundled when $\phi^c(j) = \phi^e(j)$.

An ownership profile is $g = (g_i)_{i \in \mathcal{M}} \in \mathcal{N}^{\mathcal{M}}$. For each target j, define the induced owners as $g^c(j) := g_{\phi^c(j)}$ and $g^e(j) := g_{\phi^e(j)}$.

Tasks. Project k succeeds if and only if all its tasks are completed: $x_{g^c(j)\,j} = \theta_j$ for all $j \in \mathcal{R}_k$, and $\bar{x}_{\ell j} = \theta_j$ for all $(\ell, j) \in \bar{\mathcal{R}}_k$. This set of tasks under an ownership profile g is

$$\mathcal{Q}_k(g) := \{(g^c(j), j) : j \in \mathcal{R}_k\} \;\cup\; \bar{\mathcal{R}}_k. \tag{5.2}$$

Expropriation. A party i can expropriate j's information if either it has inalienable means of expropriation ($i \notin \mathcal{E}$) or it owns the asset required to do so ($i \in \mathcal{E}$ and $i = g^e(j)$). Thus, the set of parties who are armed *against target* j is $\mathrm{Arm}_j(g) := \{i \in \mathcal{N} : i = g^e(j) \text{ or } i \notin \mathcal{E}\}$. Define the target-specific safety threshold (and set $\tau_{ii}^{\mathrm{safe}}(g) := 1$):

$$\tau_{ij}^{\mathrm{safe}}(g) \;:=\; \begin{cases} \tau_{ij}^{\star} & \text{if } i \in \mathrm{Arm}_j(g) \text{ and } i \neq j, \\ 1 & \text{otherwise.} \end{cases} \tag{5.3}$$

Total Surplus. Under Assumptions 1 and 2, the maximum attainable total surplus is

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g),$$

where $\mathcal{Q}_k(g)$ and $\tau_{ij}^{\mathrm{safe}}(g)$ are defined in (5.2) and (5.3), respectively.

Since the baseline model corresponds to $|\mathcal{M}| = 1$ and $\phi^c \equiv \phi^e$, unbundling control rights across multiple assets weakly increases maximum attainable total surplus. The next proposition shows conditions under which such unbndling can achieve the first best.

Proposition 4 (First best under separable control rights)**.** *The first-best surplus V_{total} is attainable in each of the following cases.*

(i) All requirements are alienable ($\bar{\mathcal{R}}_k = \varnothing$ for all k), and distinct requirements are assigned to distinct assets (ϕ^c is injective).

(ii) All expropriation actions are asset-dependent ($\mathcal{E} = \mathcal{N}$), and distinct targets are assigned to distinct assets (ϕ^e is injective).

When cooperation requires only alienable actions, and ownership can be arranged so that each action that depends on party i's information is controlled by party i, then no private information needs to be disclosed. When expropriation requires access to assets and ownership

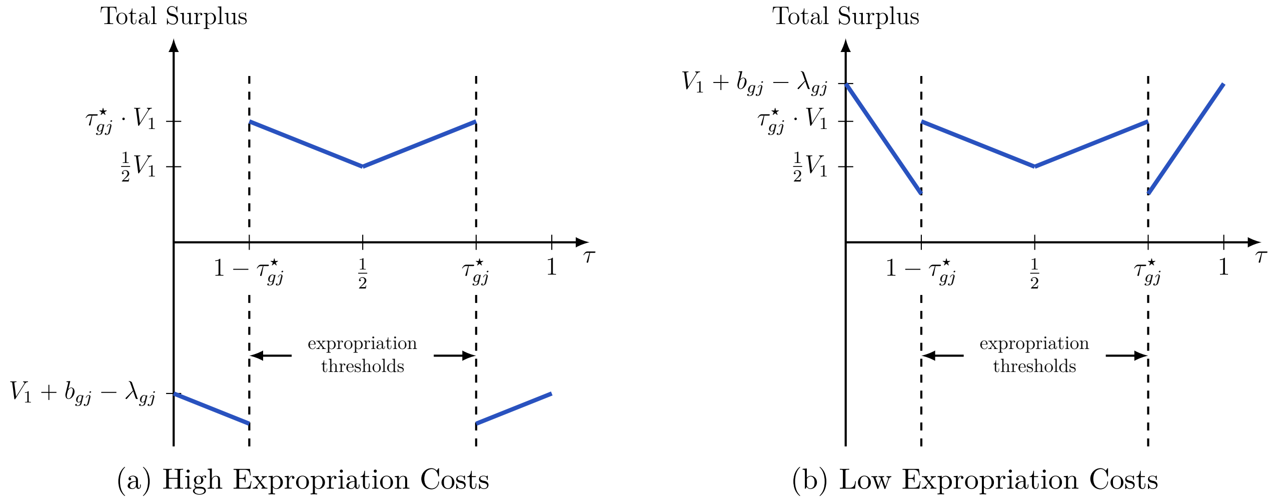

Figure 5: Optimal Disclosure in Example 3 Under Low and High Expropriation Costs

Both panels show total expected surplus as a function of the owner's belief that the non-owner's state is $\theta_j = 1$.

can be arranged so that each party controls the asset needed to expropriate *its own* information, then parties can block all expropriation attempts.

5.5 Optimal Decision Rule Under Arbitrary Expropriation Costs

The baseline model assumes successful expropriation is prohibitively costly, $\lambda_{ij} - b_{ij} > V_{\text{total}}$, so the mediator always prefers to deter expropriation. When expropriation is costly but not prohibitive, it may be optimal to tolerate some expropriation in order to raise the probability of success. The optimal ownership structure may then depend on the absolute cost of successful expropriation λ_{ij}, not just on the temptation to expropriate $\tau^\star_{ij}$.

To build intuition, consider Example 3 but with arbitrary expropriation costs: two parties $\mathcal{N} = \{1, 2\}$ and a single project $\mathcal{K} = \{1\}$; the project requires only alienable actions ($\mathcal{R}_1 = \{1, 2\}$, $\bar{\mathcal{R}}_1 = \varnothing$), and each party can expropriate ($\mathcal{E} = \varnothing$). Since the non-owner does not participate in the project, only the owner must learn the non-owner's state.

The optimal decision rule takes one of two forms, as illustrated in Figure 5: either partial disclosure at the safety thresholds when expropriation costs are high or full disclosure when expropriation costs are low. The maximum total surplus under $g \in \mathcal{N}$ is therefore

$$TS(g) = \max\{\tau^\star_{gj} \cdot V_1,\ V_1 - (\lambda_{gj} - b_{gj})\}, \text{ where } j \neq g.$$

In the baseline model, $\lambda_{gj} - b_{gj}$ is large for both $g \in \{1, 2\}$, so the first term always dominates. The optimal owner is then the party with the lower temptation to expropriate (higher $\tau^\star_{gj}$). Under arbitrary costs, the second term can dominate. Suppose party 1 has a higher temptation to expropriate ($\tau^\star_{12} < \tau^\star_{21}$) but cheaper expropriation ($\lambda_{12} - b_{12} < V_1 < \lambda_{21} - b_{21}$). If $\lambda_{12} - b_{12}$ is sufficiently close to zero, assigning the asset to party 1 is optimal: full disclosure achieves near-first-best surplus at a small expropriation cost. Appendix A.7 extends this argument to settings with only one armed party who must receive others' information, covering all our applications.

6 Conclusion

Asset ownership shapes knowledge sharing when the use of knowledge is non-contractible. This paper explores several implications of this argument. We conclude by suggesting some avenues for future research.

First, our model takes a step toward studying how transactions across firms are shaped by their internal structures. In particular, we study conditions under which firms are willing to share knowledge with an independent supplier but would be reluctant to do so if that supplier was a subsidiary of the competitor. A commonly proposed solution to this problem is an internal information barrier or what is often referred to as a Chinese wall. These safeguards matter in settings in which leaks are possible: settings with multiple rounds of communication that lie beyond the static information-design approach taken in this model.

Second, in our setting, parties commit to a mediated contract before observing their private information. If contracting is instead initiated by an informed party, the contract offer itself might convey information. In standard economic models, private information is usually represented as the realization of a random variable observed by some parties but not others, drawn from a common-knowledge distribution. In such settings, information can in principle be concealed through the design of the mechanism — an insight known as the *inscrutability principle* (Myerson, 1983).

The inscrutability logic can fail when information is "eye-opening," in the sense that it changes how parties perceive the relevant state space or feasible actions (Tirole, 2009). Then the mere act of specifying previously unforeseen contingencies expands the other party's awareness and shifts beliefs. Even if parties can contractually rule out expropriation, the disclosure required to initiate contracting changes outside options relative to no contracting, so the initial ownership structure can shape knowledge flows even when expropriation is contractible. Formalizing this setting likely requires an alternative model of information (e.g., Akerlof, Holden, and Li, 2025) and is a promising direction for future research.

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A Appendix

A.1 Proof of Proposition 1

Recall that $\mathcal{R}_k$ and $\bar{\mathcal{R}}_k$ are alienable and inalienable requirements, respectively, and $\mathcal{Q}_k(g) := \{(g,j) : j \in \mathcal{R}_k\} \cup \bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of tasks under an ownership structure $g \in \mathcal{N}$. Proposition 1 states that the maximum attainable total surplus under $g \in \mathcal{N}$ is

$$TS(g) = \sum_{k\in\mathcal{K}} V_k \cdot \left(\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right). \tag{A.1}$$

Moreover, there exists an optimal decision rule that is *nested*. Let $\xi \sim \text{Unif}[0,1]$ be a common randomization device. For each (ξ, θ, g), define an action profile $a(\xi, \theta, g) = (x, \bar{x}, z, \bar{z})$ as follows:

$$z_{ij} = 0 \;\text{ and }\; \bar{z}_{ij} = 0 \qquad \text{for all } i \in \mathcal{N},\; j \neq i, \tag{A.2}$$

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g), \\ -\theta_j & \text{if } \xi > \tau_{ij}^{\text{safe}}(g), \end{cases} \qquad \text{for all } i, j \in \mathcal{N}. \tag{A.3}$$

Define $\sigma^g(\cdot \mid \theta) \in \Delta(\mathcal{A})$ as the distribution induced by $\xi \sim \text{Unif}[0,1]$ and $a = a(\xi, \theta; g)$; that is,

$$\sigma^g(a \mid \theta) = \mathbb{P}_\xi\big[a(\xi, \theta; g) = a\big].$$

We prove the proposition in two steps. Lemma 1 shows that this decision rule is obedient and achieves $TS(g)$. Lemma 2 shows that for any obedient decision rule, the expected total surplus does not exceed $TS(g)$.

Lemma 1 (Nested rule is obedient and achieves $TS(g)$)**.** *The decision rule σ^g is obedient under an ownership structure $g \in \mathcal{N}$. Moreover, each project k succeeds with probability $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$, and hence*

$$\mathbb{E}_\psi\Big[TS(a, g; \theta)\Big] = \sum_{k\in\mathcal{K}} V_k \cdot \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = TS(g),$$

where $\psi(\theta, a) = p(\theta)\sigma^g(a \mid \theta)$ is the induced distribution over actions and states.

Proof. We verify obedience and compute the induced success probabilities.

Posteriors. First, we show that under decision rule σ^g, party i assigns probability $\tau_{ij}^{\text{safe}}(g)$ to the event that its recommendation about target j is correct. Party i observes (θ_i, a_i) before choosing actions. The expropriation recommendations in (A.2) are constant and uninformative. Since $x_{ij} = \bar{x}_{ij}$ for all i, j in (A.3), for notational convenience we will keep track of the vector $x_i := (x_{i1}, \ldots, x_{iN})$ only.

Fix i and write $x_{i,-i} := (x_{ij})_{j \neq i}$. For each $j \neq i$, let $s_{ij}(\xi) \in \{-1, 1\}$ be a deterministic sign defined by $s_{ij}(\xi) = 1$ if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ and $s_{ij}(\xi) = -1$ otherwise, so that $x_{ij} = s_{ij}(\xi) \cdot \theta_j$. For any vector $v \in \{-1, 1\}^{N-1}$ and any $\xi \in [0, 1]$, independence and uniformity of $(\theta_j)_{j \neq i}$ imply

$$\begin{aligned}\mathbb{P}_\psi(x_{i,-i} = v \mid \xi) &= \mathbb{P}_\psi\big(s_{ij}(\xi) \cdot \theta_j = v_j \ \forall j \neq i \,\big|\, \xi\big) \\ &= \prod_{j \neq i} \mathbb{P}_\psi\big(\theta_j = s_{ij}(\xi) \cdot v_j\big) = \prod_{j \neq i} \tfrac{1}{2} = 2^{-(N-1)}.\end{aligned} \tag{A.4}$$

This conditional probability does *not* depend on ξ. Hence $x_{i,-i}$ is independent of ξ. Moreover, since θ_i is independent of $(\xi, x_{i,-i})$, ξ is also independent of $(\theta_i, x_{i,-i})$. Equivalently, Bayes' rule yields that for any $v \in \{-1, 1\}^{N-1}$, the conditional density of $\xi \mid x_{i,-i}$ is

$$f_{\xi \mid x_{i,-i}}(\xi \mid v) = \frac{\mathbb{P}_\psi(x_{i,-i} = v \mid \xi)\, f_\xi(\xi)}{\mathbb{P}_\psi(x_{i,-i} = v)} = \frac{2^{-(N-1)} \cdot 1}{2^{-(N-1)}} = 1,$$

so $\xi \mid x_{i,-i}$ is uniform on $[0, 1]$. Since θ_i is independent of $(\xi, x_{i,-i})$, we also have $f_{\xi \mid \theta_i, x_{i,-i}}(\cdot \mid \theta_i, x_{i,-i}) := 1$.

Fix $j \neq i$. Under the nested rule, $\theta_j = x_{ij}$ if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$. Therefore, using iterated expectations and the posterior in (A.4),

$$\begin{aligned}\mathbb{P}_\psi(\theta_j = x_{ij} \mid \theta_i, x_{i,-i}) &= \mathbb{P}_\psi(\xi \leqslant \tau_{ij}^{\text{safe}}(g) \mid \theta_i, x_{i,-i}) \\ &= \int_0^{\tau_{ij}^{\text{safe}}(g)} f_{\xi \mid \theta_i, x_{i,-i}}(\xi \mid \theta_i, x_{i,-i})\, d\xi = \int_0^{\tau_{ij}^{\text{safe}}(g)} 1\, d\xi = \tau_{ij}^{\text{safe}}(g).\end{aligned}$$

No expropriation. Second, we show that decision rule σ^g is obedient in expropriation actions. The decision rule always prescribes no expropriation (A.2). If i is disarmed, then every effective expropriation action is 0 and there is no deviation. If i is armed, then $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^\star$ by definition. Conditional on (θ_i, a_i), the maximum success probability from attempting expropriation against any target is therefore at most $\tau_{ij}^\star$, implying that the expected private gain from expropriation is non-positive. Thus, obeying (A.2) is optimal for each party.

Obedient Cooperation. Third, we show that decision rule σ^g is obedient in cooperative actions. Consider deviations that change i's cooperative actions $(x_i, \bar{x}_i)$. For any project k, let $\mathcal{J}_{i,k}(g)$ denote the set of indices $j \neq i$ such that $(i, j) \in \mathcal{Q}_k(g)$, that is, the states of other parties that i must match (via either x_{ij} if $i = g$ and $j \in \mathcal{R}_k$, or via $\bar{x}_{ij}$ if $(i, j) \in \bar{\mathcal{R}}_k$). If $\mathcal{J}_{i,k}(g) = \varnothing$, then i's cooperative actions do not affect project k.

Fix a project k where $\mathcal{J}_{i,k}(g) \neq \varnothing$. Under the nested rule, if all parties follow their recommendations, project k succeeds if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ for all required tasks (i,j), which is equivalent to:

$$\xi \leqslant \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g). \tag{A.5}$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$ for all pairs, this probability is strictly greater than 1/2.

Now suppose party i deviates from its recommendation for a subset of targets $\tilde{\mathcal{J}}_{i,k}(g) \subseteq \mathcal{J}_{i,k}(g)$. Party i successfully matches the states if and only if the recommended actions were correct for the non-deviated targets ($j \notin \tilde{\mathcal{J}}_{i,k}(g)$) and incorrect for the deviated targets ($j \in \tilde{\mathcal{J}}_{i,k}(g)$). In terms of the randomization variable ξ, this condition requires:

$$\xi \in \Big(\max_{j\in\tilde{\mathcal{J}}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g), \min_{j\in J_{i,k}(g)\setminus\tilde{\mathcal{J}}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g) \Big], \quad \text{with the convention } \min\varnothing := 1.$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$, the lower bound of this interval is strictly greater than 1/2. Therefore, the length of the interval — and thus the success probability — is strictly less than 1/2. Consequently, any deviation yields a strictly lower probability of success than obedience.

Total surplus. Multiplying the probability of the event in (A.5) by V_k and summing over all projects yields the stated expression for $TS(g)$. $\square$

Obedient joint distributions. The next lemma establishes an upper bound on total surplus that applies to any obedient decision rule. To state it in sufficient generality, we first define obedience on joint distributions over states and actions. A joint distribution $\psi \in \Delta(\mathcal{A} \times \Theta)$ is *obedient under an ownership structure* $g \in \mathcal{N}$ if, for every $i \in \mathcal{N}$, $\theta_i \in \Theta_i$, $a_i \in \mathcal{A}_i$, and $a_i' \in \mathcal{A}_i$,

$$\sum_{\theta_{-i}\in\Theta_{-i}} \sum_{a_{-i}\in\mathcal{A}_{-i}} \psi(a_i, a_{-i}, \theta_i, \theta_{-i}) \Big[u_i((a_i, a_{-i}), g; \theta) - u_i((a_i', a_{-i}), g; \theta) \Big] \geqslant 0. \tag{A.6}$$

The obedience condition in (3.1) is a special case of (A.6): any obedient σ induces an obedient joint distribution $\psi(a, \theta) = p(\theta)\sigma(a \mid \theta)$.

Lemma 2 (Upper bound)**.** *Fix an ownership structure $g \in \mathcal{N}$. For any obedient joint distribution $\psi \in \Delta(\mathcal{A} \times \Theta)$,*

$$\mathbb{E}_{\psi}[TS(a, g; \theta)] \leqslant \sum_{k\in\mathcal{K}} V_k \cdot \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g).$$

Proof. For each task $(i,j) \in \mathcal{Q}_k(g)$ define the event that this task is completed:

$$C_{k,(i,j)} := \begin{cases} \{(\theta,a) : x_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \text{ and } (i,j) \notin \bar{\mathcal{R}}_k, \\ \{(\theta,a) : \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) \in \bar{\mathcal{R}}_k \text{ and } (i,j) \notin \{(g,j) : j \in \mathcal{R}_k\}, \\ \{(\theta,a) : x_{ij} = \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \cap \bar{\mathcal{R}}_k. \end{cases} \tag{A.7}$$

Let E_k denote the event that project k succeeds:

$$E_k := \bigcap_{(i,j) \in \mathcal{Q}_k(g)} C_{k,(i,j)}.$$

We write total surplus as

$$TS(a,g;\theta) = TS^{\text{coop}}(a,g;\theta) + TS^{\text{exp}}(a,g;\theta),$$

where $TS^{\text{coop}}(a,g;\theta) = \sum_{k \in \mathcal{K}} V_k \mathbf{1}\{E_k\}$, and TS^{exp} is the sum of all expropriation terms. We now derive an upper bound on each of these terms.

Upper Bound on Task-Completion Probabilities. Let $r_{ij} := \mathbb{P}_\psi(\zeta_{ij} \neq 0)$ be the probability that party i attempts to expropriate party j's information, $j \neq i$. If $i \notin \text{Arm}(g)$, then party i cannot engage in expropriation and $r_{ij} = 0$.

Decompose $\mathbb{P}_\psi(C_{k,(i,j)})$ according to whether i attempts to expropriate j's information:

$$\mathbb{P}_\psi(C_{k,(i,j)}) = \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) + \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0).$$

On the event $\{\zeta_{ij} = 0\}$, obedience implies that attempting expropriation is not strictly profitable for party i. Therefore, conditional on (θ_i, a_i), the probability that any task $(i,j) \in \mathcal{Q}_k(g)$ is completed is at most $\tau_{ij}^{\text{safe}}(g)$. Thus,

$$\mathbb{P}_\psi\left(C_{k,(i,j)} \mid \zeta_{ij} = 0\right) \leqslant \tau_{ij}^{\text{safe}}(g) \Rightarrow \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) \leqslant \tau_{ij}^{\text{safe}}(g)\mathbb{P}_\psi(\zeta_{ij} = 0) = \tau_{ij}^{\text{safe}}(g)(1 - r_{ij}).$$

In addition,

$$\mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0) \leqslant \mathbb{P}_\psi(\zeta_{ij} \neq 0) = r_{ij}.$$

Combining these two inequalities, we get

$$\mathbb{P}_\psi(C_{k,(i,j)}) \leqslant \tau_{ij}^{\text{safe}}(g)(1 - r_{ij}) + r_{ij}.$$

Upper bound on cooperative surplus. Fix a project $k \in \mathcal{K}$. If $\mathcal{Q}_k(g)$ contains only trivial tasks of the form (i,i), then $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g) = 1$, so $\mathbb{P}_\psi(E_k) \leqslant \min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g)$.

Fix any project k with non-trivial tasks: $(i,j) \in \mathcal{Q}_k(g)$ with $i \neq j$. Pick one bottleneck task:

$$(i_k, j_k) \in \arg\min_{(i,j)\in\mathcal{Q}_k(g)} \tau^{\text{safe}}_{ij}(g) \quad \text{with } j_k \neq i_k.$$

Since $E_k \subseteq\ C_{k,(i_k,j_k)}$, we have

$$\mathbb{P}_\psi(E_k) \leqslant \mathbb{P}_\psi(C_{k,(i,j)}) \leqslant \tau^{\text{safe}}_{i_k j_k}(g) + \big(1 - \tau^{\text{safe}}_{i_k j_k}(g)\big) r_{i_k j_k},$$

Multiplying by V_k and summing over projects yields

$$\mathbb{E}_\psi[TS^{\text{coop}}] \leqslant TS(g) + \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} \big(1 - \tau^{\text{safe}}_{ij}(g)\big)\, V^{\min}_{ij}\, r_{ij}, \tag{A.8}$$

where

$$V^{\min}_{ij} := \sum_{k:\ (i_k,j_k)=(i,j)} V_k.$$

Upper bound on expropriation surplus. Fix a pair (i,j) with $i \in \text{Arm}(g)$ and $j \neq i$. On the event $\{\zeta_{ij} \neq 0\}$, obedience implies that attempting expropriation is (weakly) optimal, hence the conditional success probability is at least the private threshold:

$$\mathbb{P}_\psi(\zeta_{ij} = \theta_j \mid \zeta_{ij} \neq 0) \ \geqslant\ \tau^\star_{ij}.$$

Therefore the expected welfare contribution from this ordered pair satisfies

$$\mathbb{E}_\psi\Big[(b_{ij} - \lambda_{ij})\,\mathbf{1}\{\zeta_{ij} = \theta_j\} - c_{ij}\,\mathbf{1}\{\zeta_{ij} = -\theta_j\}\Big] \leqslant \mathbb{E}_\psi\Big[(b_{ij} - \lambda_{ij})\,\mathbf{1}\{\zeta_{ij} = \theta_j\}\Big] \leqslant (b_{ij} - \lambda_{ij})\,\tau^\star_{ij}\, r_{ij},$$

where the first inequality follows from $-c_{ij} < 0$, and the second inequality follows from the fact that the probability of successful expropriation should be at least $\tau^\star_{ij}$. Summing over (i,j) gives

$$\mathbb{E}_\psi[TS^{\text{exp}}] \leqslant \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} \tau^\star_{ij}\,(b_{ij} - \lambda_{ij})\, r_{ij}. \tag{A.9}$$

Upper bound on total surplus. Combining (A.8) and (A.9) gives

$$\begin{aligned}
\mathbb{E}_\psi[TS(a,g;\theta)] &\leqslant TS(g) + \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[(1-\tau^\star_{ij})V^{\min}_{ij} + \tau^\star_{ij}(b_{ij} - \lambda_{ij})\Big] \\
&= TS(g) - \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[\underbrace{\tau^\star_{ij}(\lambda_{ij} - b_{ij}) - \big(1-\tau^\star_{ij}\big)V^{\min}_{ij}}_{>0}\Big].
\end{aligned}$$

By Assumption 1, $\lambda_{ij} - b_{ij} > V_{\text{total}} \geqslant V_{ij}^{\min}$. Also $b_{ij} < c_{ij}$ implies $\tau_{ij}^\star = \frac{c_{ij}}{b_{ij}+c_{ij}} > \frac{1}{2}$, hence $\tau_{ij}^\star > (1-\tau_{ij}^\star)$. Therefore, for every $i \in \text{Arm}(g)$,

$$\tau_{ij}^\star(\lambda_{ij} - b_{ij}) \;>\; \tau_{ij}^\star V_{ij}^{\min} \;\geqslant\; (1-\tau_{ij}^\star)V_{ij}^{\min},$$

so the bracketed term is strictly positive. Therefore, total surplus for any obedient decision rule is weakly less than $TS(g)$. $\square$

Conclusion. Lemma 1 shows $TS(g)$ is attainable by the nested no-expropriation rule. Lemma 2 shows no obedient rule can exceed it under Assumptions 1 and 2. This completes the proof of Proposition 1.

A.2 Proofs of Corollaries in Section 3.2

A.2.1 Corollary 1

Proof. Recall that $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^\star$ if $i \neq j$ and $i \in \text{Arm}(g)$, and $\tau_{ij}^{\text{safe}}(g) = 1$ otherwise. Hence, for each project $k \in \mathcal{K}$,

$$1 - \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = \max_{i\neq j} \left\{ \mathbf{1}\{(i,j) \in \mathcal{Q}_k(g)\} \cdot \mathbf{1}\{i \in \text{Arm}(g)\} \cdot \left(1 - \tau_{ij}^\star\right) \right\}.$$

Substituting this identity into the expression for $TS(g)$ from (3.2) yields (3.3).

Since $b_{ij} > 0$, we have $1 - \tau_{ij}^\star > 0$ for all $i \neq j$. Therefore, $TS(g) = V_{\text{total}}$ if and only if, for every $k \in \mathcal{K}$ and $i \neq j$, $\mathbf{1}\{(i,j) \in \mathcal{Q}_k(g)\} \cdot \mathbf{1}\{i \in \text{Arm}(g)\} = 0$, that is, no armed party is required to match another party's state for any project. $\square$

A.2.2 Corollary 2

Proof. Let $TS(g;h)$ denote the total surplus in (3.3) with $\tau_{ij}^\star$ replaced by $\tau_{ij}^\star(h) := \frac{c_{ij}}{b_{ij}(h)+c_{ij}}$, and let $\mathcal{G}_{\mathcal{Z}}^0$ denote the set of $\mathcal{Z}$-insulated ownership structures. Raising h weakly lowers $\tau_{ij}^\star(h)$ on the pairs in $\mathcal{Z}$ and leaves all others unchanged, so $TS(g;h)$ is weakly decreasing in h for every g; for $g \in \mathcal{G}_{\mathcal{Z}}^0$ those pairs never enter the surplus, so $TS(g;h)$ is constant in h.

By assumption, some $g^* \in \mathcal{G}_{\mathcal{Z}}^0$ is optimal at $h = \bar{h}$, so $TS(g^*;\bar{h}) \geqslant TS(g;\bar{h})$ for all g. For every g and every $h \geqslant \bar{h}$, $TS(g^*;h) = TS(g^*;\bar{h}) \geqslant TS(g;\bar{h}) \geqslant TS(g;h)$, using $g^* \in \mathcal{G}_{\mathcal{Z}}^0$ for the equality and the monotonicity of $TS(g;\cdot)$ for the last inequality. Hence, g^* remains optimal for all $h \geqslant \bar{h}$. $\square$

A.2.3 Corollary 3

Proof. If $\Gamma_\ell = \mathcal{N}$, the result is immediate. Suppose $\Gamma_\ell \neq \mathcal{N}$. Let g^*_ℓ be an ownership structure in Γ_ℓ that minimizes the value-weighted loss in projects other than ℓ:

$$g^*_\ell \in \arg\min_{g' \in \Gamma_\ell} \sum_{k \neq \ell} V_k \cdot \mathrm{Loss}_k(g').$$

For any $g \notin \Gamma_\ell$,

$$TS(g^*_\ell) - TS(g) = V_\ell \underbrace{\left[\mathrm{Loss}_\ell(g) - \mathrm{Loss}_\ell(g^*_\ell)\right]}_{>0} + \sum_{k \neq \ell} V_k \left[\mathrm{Loss}_k(g) - \mathrm{Loss}_k(g^*_\ell)\right].$$

The first term grows linearly in V_ℓ with a strictly positive coefficient, while the second term is independent of V_ℓ. Hence, for each $g \notin \Gamma_\ell$, there exists a finite $\bar{V}_\ell(g)$ such that $TS(g^*_\ell) > TS(g)$ for all $V_\ell > \bar{V}_\ell(g)$. Taking $\bar{V}_\ell := \max_{g \notin \Gamma_\ell} \bar{V}_\ell(g)$, we have $TS(g^*_\ell) > TS(g)$ for all $g \notin \Gamma_\ell$ when $V_\ell > \bar{V}_\ell$. By construction, $TS(g^*_\ell) \geqslant TS(g')$ for all $g' \in \Gamma_\ell$, so g^*_ℓ is optimal and no ownership structure outside Γ_ℓ can be optimal. $\square$

A.3 Implementation Through a Mediated Contract

This appendix proves the results stated in Section 5.1. We first analyze the communication game and prove Proposition 2. We then analyze the contracting game and show how the outcome from the baseline solution concept can be supported as a Perfect Bayesian Equilibrium (PBE).

A.3.1 Communication Game and Proof of Proposition 2

Incentive Constraints. Fix a direct mediated mechanism (φ, t). When all parties report truthfully and follow their recommendations, party i's interim expected utility is

$$U_i(\varphi, t \mid \theta_i) \;:=\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a \in \mathcal{A}} \sum_{g \in \mathcal{N}} \varphi(a, g \mid \theta_i, \theta_{-i}) \Big[u_i(a, g; \theta_i, \theta_{-i}) + t_i\big((a, g), (\theta_i, \theta_{-i})\big) \Big].$$

Parties are not compelled to follow recommendations; a (pure) deviation for party i is a function $\delta_i : D_i \to \mathcal{A}_i$ mapping the available information at the action stage into an action plan, where $D_i := \mathcal{N} \times \mathcal{T} \times \mathcal{A}_i \times \Theta_i$, and party i observes $d_i := (g, t', a_i, \theta_i) \in D_i$. For any alternative report $\theta'_i \in \Theta_i$ and any deviation function $\delta_i : D_i \to \mathcal{A}_i$, define the *double-deviation* payoff

$$U^\star_i(\varphi, t, \delta_i, \theta'_i \mid \theta_i) \;:=\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a \in \mathcal{A}} \sum_{g \in \mathcal{N}} \varphi(a, g \mid \theta'_i, \theta_{-i}) \Big[u_i\big((\delta_i(g, t', a_i, \theta_i), a_{-i}), g; \theta_i, \theta_{-i}\big) + t_i\big((a, g), (\theta'_i, \theta_{-i})\big) \Big],$$

where $t' := t((a,g),(\theta_i',\theta_{-i}))$ is the transfer vector implemented after the report (θ_i',θ_{-i}).

The contract (φ,t) is *incentive compatible* if, for every $i \in \mathcal{N}$,

$$U_i(\varphi,t \mid \theta_i) \;\geqslant\; U_i^\star(\varphi,t,\delta_i,\theta_i' \mid \theta_i) \qquad \text{for all } \theta_i,\theta_i' \in \Theta_i \text{ and all } \delta_i : D_i \to \mathcal{A}_i. \tag{A.10}$$

This condition subsumes truth-telling (take $\delta_i(d_i) = a_i$) and obedience (take $\theta_i' = \theta_i$) as special cases.

Communication-Game Problem. The maximum attainable expected total surplus in the communication game is

$$\begin{aligned} \max_{\varphi,t} \quad & \sum_{\theta\in\Theta} p(\theta) \sum_{g\in\mathcal{N}} \sum_{a\in\mathcal{A}} \varphi(a,g \mid \theta)\, TS(a,g;\theta) \\ \text{subject to} \quad & \text{(A.10)}, \\ & \varphi(a,g \mid \theta) \geqslant 0, \qquad \sum_{a,g} \varphi(a,g \mid \theta) = 1 \qquad \text{for all } \theta \in \Theta, \\ & \sum_{i\in\mathcal{N}} t_i(a,g,\theta) = 0 \qquad \text{for all } (a,g,\theta) \in \mathcal{A}\times\mathcal{N}\times\Theta. \end{aligned} \tag{A.11}$$

Relaxed Problem: Obedience Only. Consider the relaxation of (A.11) obtained by dropping the truth-telling and the double-deviation constraints. In this relaxation, we evaluate incentives under the *truthful* report profile and impose only obedience of the recommended non-contractible actions.

Because transfers are implemented before actions are chosen and do not depend on realized non-contractible actions, they cancel from the obedience constraint when comparing a recommended action to a deviation. The obedience constraints are: for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every $g \in \mathcal{N}$, every recommended action plan $a_i \in \mathcal{A}_i$, and every deviation $a_i' \in \mathcal{A}_i$,

$$\sum_{\theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i}} \varphi\left(a_i,a_{-i},g \mid \theta_i,\theta_{-i}\right)\left[u_i\left(a_i,a_{-i},g;\theta\right) - u_i\left(a_i',a_{-i},g;\theta\right)\right] \geqslant 0. \tag{A.12}$$

Thus, the relaxed problem is

$$\max_{\varphi} \quad \sum_{\theta\in\Theta} p(\theta) \sum_{g\in\mathcal{N}} \sum_{a\in\mathcal{A}} \varphi(a,g \mid \theta)\, TS(a,g;\theta) \tag{A.13}$$
$$\text{subject to} \quad \text{(A.12)}, \qquad \varphi(a,g \mid \theta) \geqslant 0, \quad \sum_{a,g} \varphi(a,g \mid \theta) = 1 \;\; \text{for all } \theta \in \Theta.$$

For any fixed ownership structure $g \in \mathcal{N}$, if we restrict attention to allocation rules that put probability one on g, then (A.13) coincides with the benchmark mediator problem (2.1) defining $TS(g)$. The following lemma shows that focusing on deterministic ownership structures is without loss of optimality.

Lemma 3 (Deterministic ownership)**.** *In the communication-game problem* (A.13)*, there exists an optimal solution with a deterministic ownership structure.*

Proof. Let φ be any feasible allocation rule for (A.13), and let $G \in \mathcal{N}$ denote the (possibly random) owner induced by φ (which is publicly observed before actions are chosen). Let $\psi(a, g, \theta) := p(\theta)\varphi(a, g \mid \theta)$ denote the joint distribution of (a, g, θ) induced by φ.

Fix any $g \in \mathcal{N}$ with $\mathbb{P}_\psi(G = g) > 0$, and define the conditional joint distribution of actions and states given $G = g$ by

$$\psi^g(a, \theta) := \mathbb{P}_\psi(a, \theta \mid G = g) = \frac{\psi(a, g, \theta)}{\mathbb{P}_\psi(G = g)}.$$

Because the obedience constraints (A.12) are imposed separately for each ownership realization g, and because parties observe $G = g$ before choosing actions, the conditional distribution ψ^g is obedient under owner g. Therefore, applying Lemma 2 with $\psi = \psi^g$,

$$\mathbb{E}_{\psi^g}\big[TS(a, g; \theta)\big] \leqslant TS(g).$$

Taking expectations over G yields

$$\mathbb{E}_\psi[TS(a, G; \theta)] = \sum_{g \in \mathcal{N}} \mathbb{P}_\psi(G = g)\, \mathbb{E}_{\psi^g}[TS(a, g; \theta)] \leqslant \sum_{g \in \mathcal{N}} \mathbb{P}_\psi(G = g)\, TS(g) \leqslant \max_{g \in \mathcal{N}} TS(g).$$

Finally, choosing deterministic ownership $G \equiv g^* \in \arg\max_{g \in \mathcal{N}} TS(g)$ and an optimal obedient decision rule that attains $TS(g^*)$ achieves the upper bound. Hence an optimal solution to (A.13) exists with deterministic ownership. □

The relaxed problem with deterministic ownership provides an upper bound for the communication-game problem (A.11) and coincides with the problem in (2.1). Next, we show that this upper bound is attainable.

Lemma 4 (Constant-transfer implementation)**.** *Let $\varphi^*(a, g \mid \theta)$ be defined as in* (5.1)*. Then, for any constant budget-balanced transfer vector $\bar{t} \in \mathcal{T}$, the direct mediated mechanism $(\varphi^*, \bar{t})$ is incentive compatible.*

Proof. Because transfers are constant, they cancel out in all incentive-compatibility constraints. Fix a party i, a true type θ_i, a report $\hat{\theta}_i \in \Theta_i$, and an action-stage strategy $\delta_i : D_i \to \mathcal{A}_i$.

Step 1: Upper Bound for a Single Task. Fix any target $j \neq i$. Suppose all parties except i truthfully report their states and obey their recommendations. Then party i receives recommendations

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j, & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g^*), \\ -\theta_j, & \text{if } \xi > \tau_{ij}^{\text{safe}}(g^*). \end{cases} \tag{A.14}$$

In Lemma 1, we proved that the posterior belief that $\theta_j = x_{ij}$ under (A.14) is equal to $\tau_{ij}^{\text{safe}}(g^*)$. Since $\tau_{ij}^{\text{safe}}(g^*) > \frac{1}{2}$, the event $\theta_j = x_{ij}$ is more likely than $\theta_j = -x_{ij}$ given d_i. Thus, the maximum achievable success probability in task (i,j) equals $\tau_{ij}^{\text{safe}}(g^*)$. Therefore, for any strategy δ_i and any report $\hat{\theta}_i$, the probability that task (i,j) in project k is completed (event $C_{k,(i,j)}$ defined in (A.7)) has the following upper bound:

$$\mathbb{P}(C_{k,(i,j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \;\leqslant\; \tau_{ij}^{\text{safe}}(g^*) \qquad \text{for all } j \neq i. \tag{A.15}$$

Fix $\ell \neq i$ and $j \neq \ell$. Since ℓ obeys its recommendations, its tasks are completed with the following probabilities:

$$\mathbb{P}(C_{k,(\ell,j)} \mid \hat{\theta}_i, \delta_i, \theta_i) = \begin{cases} \tau_{\ell j}^{\text{safe}}(g^*), & \text{if } j \neq i \text{ (since } \hat{\theta}_j = \theta_j), \\ \tau_{\ell i}^{\text{safe}}(g^*), & \text{if } j = i,\ \hat{\theta}_i = \theta_i, \\ 1 - \tau_{\ell i}^{\text{safe}}(g^*), & \text{if } j = i,\ \hat{\theta}_i = -\theta_i. \end{cases}$$

Furthermore, since $\tau_{\ell i}^{\text{safe}}(g^*) \geqslant \frac{1}{2}$, in all cases we have

$$\mathbb{P}(C_{k,(\ell,j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \tau_{\ell j}^{\text{safe}}(g^*). \tag{A.16}$$

Step 2: Upper Bound on Project Success. For each project k, success requires all required tasks to be completed: $E_k = \bigcap_{(\ell,j)\in\mathcal{Q}_k(g^*)} C_{k,(\ell,j)}$. Hence, for any deviation $\hat{\theta}_i$ and δ_i,

$$\mathbb{P}(E_k \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \mathbb{P}(C_{k,(\ell,j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \tau_{\ell j}^{\text{safe}}(g^*).$$

Under truthful reporting and obedience, each required task (ℓ, j) is completed if and only if $\xi \leqslant \tau_{\ell j}^{\text{safe}}(g^*)$, so (because the same ξ is shared across tasks)

$$\mathbb{P}(E_k \mid \text{truth, obey}, \theta_i) = \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \tau_{\ell j}^{\text{safe}}(g^*).$$

That is, truthful reporting and obedience attain the upper bound for every project.

Since $u_i^c = \sum_{k\in\mathcal{K}} \alpha_{ik} V_k \,\mathbf{1}\{E_k\}$ with $\alpha_{ik} V_k \geqslant 0$, maximizing each $\mathbb{P}(E_k)$ implies that truthful reporting and obedience maximize i's expected cooperative payoff.

No Expropriation. Under σ^* the posterior belief that $\theta_j = x_{ij}$ is capped at $\tau_{ij}^{\text{safe}}(g^*)$, so by the definition of the safety threshold (see Lemma 1), any expropriation attempt yields a weakly lower expected payoff than $\zeta_{ij} = 0$. The expropriation payoff is unaffected by i's report because i's information about θ_{-i} is independent of $\hat{\theta}_i$.

Conclusion. Combining the cooperative and expropriation parts yields the double-deviation constraints (A.10), and therefore the mechanism is incentive-compatible. □

Proposition 2. Lemma 3 shows that $TS(g^*)$ provides an upper bound on the maximum surplus in the communication game. Lemma 4 shows that this surplus is attainable. Together, these results complete the proof of Proposition 2.

A.3.2 Contracting Game

Let (a^0, g_0) denote the disagreement outcome, where $x_{ii} = \bar{x}_{ii} = \theta_i$ for all i, $x_{ij} = \bar{x}_{ij} = 1$ for all $j \neq i$, and $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$. Let

$$\underline{U}_i := \sum_{\theta \in \Theta} p(\theta)\, u_i(a^0, g_0; \theta)$$

denote party i's ex ante disagreement payoff. Let

$$\bar{U}_i^* := \sum_{\theta \in \Theta} p(\theta) \sum_{a \in \mathcal{A}} \sigma^*(a \mid \theta)\, u_i(a, g^*; \theta)$$

denote party i's expected non-transfer payoff under the optimal decision rule. Define the constant transfer vector $t^* \in \mathcal{T}$ by

$$t_i^* = \underline{U}_i - \bar{U}_i^* \quad (i \neq 1), \qquad t_1^* = -\sum_{i \neq 1} t_i^*.$$

Let (φ^*, t^*) denote the associated direct mediated contract, where φ^* is defined in (5.1).

Proposition 5 (PBE implementation of the optimal contract)**.** *There exists a Perfect Bayesian Equilibrium of the contracting game in which:*

1. *Party 1 offers the contract (φ^*, t^*).*
2. *Every party $i \neq 1$ accepts.*
3. *Conditional on acceptance of (φ^*, t^*), every party reports truthfully and obeys its recommendation on the equilibrium path.*

The induced on-path joint distribution over ownership, states, and actions is

$$p(\theta)\,\mathbf{1}\{g = g^*\}\sigma^*(a \mid \theta),$$

and total surplus equals $\max_{g \in \mathcal{N}} TS(g)$.

Proof. For each (φ, t) among feasible direct mediated contracts, the continuation (communication) game induced by (φ, t) is a finite extensive-form game with perfect recall. Hence, it has a sequential equilibrium (Kreps and Wilson, 1982), and therefore at least one PBE. For each feasible contract $(\varphi, t) \neq (\varphi^*, t^*)$, fix one continuation PBE of the induced communication game, and let $W_i(\varphi, t)$ denote party i's ex ante payoff in that continuation equilibrium. For the optimal contract (φ^*, t^*), choose the truthful-reporting and obedient continuation Bayesian Nash equilibrium from Lemma 4, and let $W_i(\varphi^*, t^*)$ be the associated ex ante payoff for party i. We will show that these strategies can be supported as a PBE of the continuation game. Finally, consider the continuation game following the rejection of a contract. The disagreement actions a^0 constitute a PBE strategy profile of this game.

In stage 1, party 1 offers (φ^*, t^*). For each $i \neq 1$, define the stage-2 acceptance rule by

$$\chi_i(\varphi, t) = \begin{cases} \text{accept}, & \text{if } W_i(\varphi, t) \geqslant \underline{U}_i, \\ \text{reject}, & \text{if } W_i(\varphi, t) < \underline{U}_i. \end{cases}$$

Because t^* was chosen so that $W_i(\varphi^*, t^*) = \underline{U}_i$ for every $i \neq 1$, and every party $i \neq 1$ accepts the offer (φ^*, t^*).

Conditional on acceptance of (φ^*, t^*), let every party report truthfully and obey its recommendation on path. By Lemma 4, this continuation strategy profile is a Bayesian Nash equilibrium of the communication game induced by (φ^*, t^*). Since the prior has full support, every report profile to the mediator occurs with positive probability. Consider any action-stage information set of party i, $d_i = (g, \tilde{t}, a_i, \theta_i) \in D_i$, that occurs with zero probability on the equilibrium path. We distinguish two cases.

First, d_i may be *report-consistent*: there exists a report profile $(\hat{\theta}_i, \theta_{-i}) \in \Theta$ and an action profile $a = (a_i, a_{-i})$ such that $\varphi^*(a, g \mid (\hat{\theta}_i, \theta_{-i})) > 0$ and $\tilde{t} = t^*((a, g), (\hat{\theta}_i, \theta_{-i}))$. At such histories, beliefs are determined by Bayes' rule conditional on d_i and on party i's information about its own report. The continuation actions of party i are then chosen as a best response under these beliefs.

Second, d_i may be *report-inconsistent*: there is no report profile $(\hat{\theta}_i, \theta_{-i}) \in \Theta$ and no action profile $a = (a_i, a_{-i})$ such that $\varphi^*(a, g \mid (\hat{\theta}_i, \theta_{-i})) > 0$ and $\tilde{t} = t^*((a, g), (\hat{\theta}_i, \theta_{-i}))$. At such histories, beliefs are unrestricted by Bayes' rule, and we assign the belief μ_i described below.

For each report-inconsistent information set d_i and each $j \neq i$, define a productive sign $\tilde{s}_{ij}(d_i) \in \{-1, 1\}$ by

$$\tilde{s}_{ij}(d_i) := \begin{cases} x_{ij}, & \text{if } x_{ij} = \bar{x}_{ij} \in \{-1, 1\}, \\ 1, & \text{otherwise.} \end{cases}$$

At d_i, prescribe the continuation actions $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$, $x_{ii} = \bar{x}_{ii} = \theta_i$, and $x_{ij} = \bar{x}_{ij} = \tilde{s}_{ij}(d_i)$ for $j \neq i$. Party i's off-path belief $\mu_i(\cdot \mid d_i)$ over θ_{-i} is generated by a latent common random variable $\xi \sim \text{Unif}[0, 1]$ according to

$$\theta_j = \begin{cases} \tilde{s}_{ij}(d_i), & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g), \\ -\tilde{s}_{ij}(d_i), & \text{if } \xi > \tau_{ij}^{\text{safe}}(g), \end{cases} \qquad \text{for every } j \neq i.$$

Moreover, party i believes that the same latent ξ governs all other parties' productive recommendations under owner g, and that all other parties follow their continuation strategies. Under these beliefs, each sign $\tilde{s}_{ij}(d_i)$ is the most likely realization of θ_j, and the posterior belief that $\theta_j = \tilde{s}_{ij}(d_i)$ is $\tau_{ij}^{\text{safe}}(g)$. Hence, no expropriation is weakly optimal, and the prescribed cooperative action plan is sequentially rational. Therefore, the Bayes Nash equilibrium from Lemma 4 can be supported as a PBE of the continuation game.

Finally, by Proposition 2, no contract can attain surplus strictly above $\max_{g \in \mathcal{N}} TS(g)$, whereas (φ^*, t^*) attains that value. If all parties accept a contract (φ, t), then

$$W_1(\varphi, t) = \sum_{i=1}^{N} W_i(\varphi, t) - \sum_{i \neq 1} W_i(\varphi, t) \leqslant \max_{g \in \mathcal{N}} TS(g) - \sum_{i \neq 1} \underline{U}_i = W_1(\varphi^*, t^*),$$

where the inequality uses Proposition 2 and the fact that acceptance implies $W_i(\varphi, t) \geqslant \underline{U}_i$ for all $i \neq 1$. If some party rejects (φ, t), party 1 receives the disagreement payoff, which is weakly below $W_1(\varphi^*, t^*)$. Hence, party 1 weakly prefers offering (φ^*, t^*). Therefore, the offer strategy, acceptance rules, continuation strategies, and belief profile constructed above define a PBE of the contracting game. □

A.4 Contractible Actions and Cooperative Payoffs

This section studies four counterfactual environments that differ in the set of contractible variables: (i) expropriation actions, (ii) cooperative actions, (iii) project shares, and (iv) both cooperative actions and project shares.

For notational convenience, decompose each party's action plan into a *cooperation block* and an *expropriation block*:

$$a_i^c := (x_i, \bar{x}_i) \in \mathcal{A}_i^c = \{-1, 1\}^{2N}, \qquad a_i^e := (z_i, \bar{z}_i) \in \mathcal{A}_i^e = \{-1, 0, 1\}^{2(N-1)},$$

so that $\mathcal{A}_i = \mathcal{A}_i^c \times \mathcal{A}_i^e$ and $a_i = (a_i^c, a_i^e)$. The mediator observes θ and draws an action profile $a = (a_i)_{i \in \mathcal{N}} \in \mathcal{A}$ from a decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$.

(i) Contractible expropriation actions. Suppose the mediator can *enforce* expropriation actions $(z, \bar{z})$, while cooperative actions $(x, \bar{x})$ remain non-contractible. Then, after receiving a recommendation $a_i = (a_i^c, a_i^e)$, party i can deviate only in the cooperation block a_i^c. Accordingly, σ is *obedient under g with contractible expropriation actions* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a_i^c, a_i^e) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i^c \in \mathcal{A}_i^c$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a_i^c, a_i^e), a_{-i}), g; \theta\big) - u_i\big(((\tilde{a}_i^c, a_i^e), a_{-i}), g; \theta\big) \Big] \geqslant 0. \tag{A.17}$$

Define the maximum attainable value with contractible expropriation actions:

$$TS^e(g) := \max_{\sigma} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.18}$$
$$\text{subject to (A.17)}, \quad \sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

(ii) Contractible cooperative actions. Suppose the mediator can enforce cooperative actions $(x, \bar{x})$, while expropriation actions $(z, \bar{z})$ remain non-contractible. Then, after receiving a recommendation $a_i = (a_i^c, a_i^e)$, party i can deviate only in the expropriation block a_i^e. Accordingly, σ is *obedient under g with contractible cooperative actions* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a_i^c, a_i^e) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i^e \in \mathcal{A}_i^e$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a_i^c, a_i^e), a_{-i}), g; \theta\big) - u_i\big(((a_i^c, \tilde{a}_i^e), a_{-i}), g; \theta\big) \Big] \geqslant 0. \tag{A.19}$$

The maximum attainable value with contractible cooperative actions is

$$TS^c(g) := \max_{\sigma} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.20}$$
$$\text{subject to (A.19)}, \quad \sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

(iii) Contractible project shares. Suppose the mediator can enforce project shares $\{\alpha_{ik}\}$, while all actions are non-contractible. The obedience constraints remain the same as in (3.1).

The maximum attainable surplus with contractible project shares is

$$TS^{\alpha}(g) := \max_{\sigma,\{\alpha_{ik}\}} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.21}$$
$$\text{subject to (3.1)}, \quad \alpha_{ik} \geqslant 0, \quad \sum_{i \in \mathcal{N}} \alpha_{ik} = 1 \text{ for all } k \in \mathcal{K},$$
$$\sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

(iv) Contractible cooperative actions and project shares. Suppose instead that the mediator can *enforce* cooperative actions $(x, \bar{x})$ and project shares $\{\alpha_{ik}\}$, while expropriation actions $(z, \bar{z})$ remain non-contractible. The obedience constraints are the same as in (A.19). The maximum attainable surplus is

$$TS^{c,\alpha}(g) := \max_{\sigma,\{\alpha_{ik}\}} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.22}$$
$$\text{subject to (A.19)}, \quad \alpha_{ik} \geqslant 0, \quad \sum_{i \in \mathcal{N}} \alpha_{ik} = 1 \text{ for all } k \in \mathcal{K},$$
$$\sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta.$$

Proposition 6 (Contractibility and attainable surplus)**.** *Fix any ownership assignment $g \in \mathcal{N}$. If expropriation actions $(z, \bar{z})$ are contractible, then*

$$TS^{e}(g) = V_{\text{total}} \qquad \textit{for every } g \in \mathcal{N}.$$

If cooperative actions $(x, \bar{x})$ and/or project shares $\{\alpha_{ik}\}_{i,k}$ are contractible, then

$$TS^{c}(g) = TS^{\alpha}(g) = TS^{c,\alpha} = TS(g) \qquad \textit{for every } g \in \mathcal{N},$$

where $TS(g)$ is the value characterized in Proposition 1.

Proof. **(i) Contractible expropriation actions:** $TS^{e}(g) = V_{\text{total}}$. Define a (deterministic) decision rule σ^{FB} as follows. For every $\theta \in \Theta$:

1. Set *all* expropriation actions to zero: $z_{ij} = 0$ and $\bar{z}_{ij} = 0$ for all i and all $j \neq i$;

2. Recommend $x_{ij} = \bar{x}_{ij} = \theta_j$ for all $i \in \mathcal{N}$ and $j \in \mathcal{N}$.

Under σ^{FB}, there is no expropriation, and every project succeeds surely. Thus the total surplus is V_{total}. Since $(z, \bar{z})$ are enforced, party i can deviate only in $a_i^c = (x_i, \bar{x}_i)$. Any deviation in a_i^c

can only decrease te probability of project success; since cooperative payoffs have nonnegative shares, such a deviation cannot increase i's payoff. Thus, σ^{FB} is obedient.

(ii)–(iv) Contractible cooperative actions and/or project shares. The proof of Lemma 2 derives an upper bound on total surplus using only obedience constraints for expropriation actions. Thus, the same upper bound applies when cooperative actions and/or project shares are contractible. The same decision rule as in Proposition 1 attains this upper bound. □

A.5 Non-Contractible Communication: Unmediated Cheap Talk

This appendix proves the results stated in Section 5.3. We first prove Proposition 3 regarding the cheap-talk game. We then show how to adapt the reasoning from the proof of Proposition 1 for the binary-state case to the continuum analogue.

A.5.1 Proof of Proposition 3

Proof. Fix an ownership structure $g \in \mathcal{N}$. We construct a PBE in which each task (i, j) is completed with probability $\tau_{ij}^{\text{safe}}(g)$, and there is no expropriation on the equilibrium path. Because each state θ_j is communicated to at most one armed party i, it suffices to separate construct a communication strategy within each task (i, j).

Consider first any task (i, j) with $i \notin \mathrm{Arm}(g)$. Since party i is disarmed, sender j can truthfully reveal θ_j to i, and task (i, j) is completed with probability $1 = \tau_{ij}^{\text{safe}}(g)$.

Success and Failure Regions. Now consider any task (i, j) with $i \in \mathrm{Arm}(g)$. To simplify notation, let $\tau^\star := \tau_{ij}^\star$ and $\delta := \delta_j$. Define the success interval and failure interval in $\mathbb{T} = [0, 1)$ by $\mathbb{T}^{\text{success}} := [0, \tau^\star]$ and $\mathbb{T}^{\text{fail}} := (\tau^\star, 1)$.

Partition of the Success Region. Let $s^\circ := \min\{2\delta, \frac{\tau^\star}{2}\}$. We partition $\mathbb{T}^{\text{success}}$ into $M := \lceil \tau^\star/(2\delta) \rceil$ contiguous intervals $S^1, \ldots, S^M$ such that

$$S^1 := [0, s^\circ), \qquad S^M := [\tau^\star - s^\circ, \tau^\star], \qquad |S^m| \leqslant 2\delta \quad \text{for every } m,$$

and $\bigcup_{m=1}^M S^m = \mathbb{T}^{\text{success}}$. Write $S^m = [\nu^{m-1}, \nu^m)$ with $\nu^0 := 0$ for $m = 1, \ldots, M-1$ and $S^m = [\nu^{m-1}, \nu^m]$ with $\nu^M := \tau^\star$. Thus, $|S^m| = \nu^m - \nu^{m-1}$.

For each interval S^m, choose an intended receiver action x^m such that

$$x^1 = \delta, \qquad x^M = \tau^\star - \delta, \qquad x^m = \frac{\nu^{m-1} + \nu^m}{2} \quad \text{for } m = 2, \ldots, M-1$$

Partition of the Failure Region. Define the linear mapping:

$$\phi(y) := \tau^\star + \frac{1-\tau^\star}{\tau^\star} \cdot y \quad \text{for } y \in \mathbb{T}^{\text{success}}.$$

Define failure intervals as follows:

$$F^1 := (\tau^\star, \phi(\nu^1)), \qquad F^M := [\phi(\nu^{M-1}), 1), \qquad F^m := \phi(S^m) \quad \text{for } m = 2, \ldots, M-1.$$

The sets $\{F^m\}_{m=1}^M$ are disjoint intervals that partition $\mathbb{T}^{\text{fail}}$, and

$$|F^m| = \frac{1-\tau^\star}{\tau^\star}|S^m|.$$

Strategies. The sender's strategy is to send message $m \in \{1, 2, \ldots, M\}$ if and only if $\theta_j \in S^m \cup F^m$. Upon receiving message m, the receiver chooses the intended cooperative action $x_{ij} = \bar{x}_{ij} = x^m$ and does not attempt to expropriate ($\zeta = \varnothing$).

Sender incentive compatibility. If $\theta_j \in S^m$, then by construction

$$\theta_j \in B(x^m, \delta) := \{\theta : d(x^m, \theta) \leqslant \delta\},$$

so the prescribed message yields task (i, j) success with probability one.

If $\theta_j \in F^m$, then θ_j lies strictly outside every receiver success interval. Under any message, the task fails:

$$B(x^{m'}, \delta) \cap \left(\mathbb{T}^{\text{fail}}\right) = \varnothing \quad \text{for all } m' \in \{1, 2, \ldots, M\}.$$

Thus, sender types in F^m are indifferent between all messages and therefore do not have a profitable deviation.

Receiver incentive compatibility. After receiving message m, the receiver's posterior is uniform on $S^m \cup F^m$. By construction of the failure intervals, the relative measure of the success interval satisfies

$$\frac{|S^m|}{|S^m| + |F^m|} = \frac{|S^m|}{|S^m| + \frac{1-\tau^\star}{\tau^\star}|S^m|} = \tau^\star.$$

Since the intended action x^m covers all S^m and none of F^m, the probability of success from choosing x^m is $\tau^\star$.

We verify that no other action $y \in \mathbb{T}$ can yield a strictly higher probability of success. The success mass covered by any action y is $|B(y, \delta) \cap (S^m \cup F^m)|$. If $B(y, \delta)$ intersects at most one of the two intervals, the covered mass is bounded by

$$|B(y,\delta) \cap (S^m \cup F^m)| \leqslant \max\{|S^m|, |F^m|\} = |S^m|,$$

where the equality holds because $\tau^\star > 1/2$ implies $|F^m| < |S^m|$. Such a deviation yields a success probability of at most $\tau^\star$, so it is weakly unprofitable.

Suppose $M \geqslant 3$. This implies $\tau^\star > 4\delta$ and therefore $|S^1| = |S^M| = 2\delta$. For any interior message $m \in \{2, \ldots, M-1\}$, the distance between S^m and F^m in either direction around the circle is bounded below by 2δ. Thus, $B(y,\delta)$ cannot simultaneously intersect both S^m and F^m, and for any interior message $m \in \{2, \ldots, M-1\}$, any action covers at most mass $|S^m|$, meaning there is no profitable deviation from x^m.

For $M \geqslant 2$, consider $m \in \{1, M\}$ and a deviation $y \in \mathbb{T}$ such that $B(y,\delta)$ simultaneously intersects both S^1 and F^1. By symmetry, it suffices to analyze $m = 1$.

Intervals S^1 and F^1 are separated by two gaps on the circle $\mathbb{T}$. First, there is the counter-clockwise gap spanning the remainder of the success region, with length $D_{ccw} = \tau^\star - |S^1|$. If $D_{ccw} \geqslant 2\delta$, then a radius-$\delta$ neighborhood cannot cover this gap and hence cannot simultaneously intersect both S^1 and F^1. If $D_{ccw} < 2\delta$, then such a neighborhood covers at most $2\delta - D_{ccw} = |S^1| - (\tau^\star - 2\delta) < |S^1|$ of $S^1 \cup F^1$, where the last inequality uses $2\delta < \tau^\star$. Therefore, there is no action $y \in \left[\delta, \tau^\star + \frac{1-\tau^\star}{\tau^\star} s^\circ - \delta\right]$ that can yield higher success probability than x^1.

Second, there is the clockwise gap spanning the remainder of the failure region, with length $D_{cw} = 1 - \tau^\star - |F^1|$. An interval of length 2δ covering this gap covers at most $2\delta - D_{cw} = 2\delta - 1 + \tau^\star + |F^1|$ of $S^1 \cup F^1$. We show that the covered mass satisfies the following inequality:

$$2\delta - 1 + \tau^\star + |F^1| \leqslant |S^1|.$$

To see this, note that because $|F^1| = \frac{1-\tau^\star}{\tau^\star}|S^1|$, the inequality can be written as

$$|S^1| \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1}. \tag{A.23}$$

By construction, $|S^1| = \min\{2\delta, \tau^\star/2\}$. If $|S^1| = \tau^\star/2$, then

$$\frac{\tau^\star}{2} \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1} \iff \frac{1}{4} \geqslant \delta,$$

which is satisfied by Assumption $2^\star$, that is, $\delta \leqslant 1/4$. If $|S^1| = 2\delta$, then

$$2\delta \geqslant \tau^\star \frac{\tau^\star + 2\delta - 1}{2\tau^\star - 1} \iff (1 - \tau^\star)(\tau^\star - 2\delta) \geqslant 0,$$

which is satisfied by Assumption $2^\star$, that is, $\tau^\star > 1/2$ and $\delta \leqslant 1/4$. Thus, the inequality (A.23) is satisfied, and no action $y \in (\tau^\star + \frac{1-\tau^\star}{\tau^\star}s^\circ - \delta, 1) \cup [0, \delta)$ that can yield a higher expected success probability than x^1.

By a symmetric geometric construction, the same result holds for $m = M$, concluding the proof of receiver incentive compatibility.

Probability of Success. In the constructed PBE, a task (i, j) with $i \in \mathrm{Arm}(g)$ is completed if θ_j is in the success region, which occurs with probability $\tau_{ij}^\star = \tau_{ij}^{\mathrm{safe}}(g)$. □

A.5.2 Deterministic Decision Rule With Continuous State and Action Space

We show that in the special case where only one armed party receives information from others — which covers Examples 3 and 4 and all applications — there exists a deterministic decision rule attaining the upper bound on total surplus.

Proposition 7. *Fix an ownership structure $g \in \mathcal{N}$. Suppose there exists a party $i \in \mathcal{N}$ such that every payoff-relevant task in $\bigcup_{k\in\mathcal{K}} \mathcal{Q}_k(g)$ is one of the following: (i) a self-task (ℓ, ℓ), (ii) a task (ℓ, j) performed by a disarmed party $\ell \notin \mathrm{Arm}(g)$, (iii) a task (i, j). There exists a deterministic obedient decision rule that attains the upper bound on total surplus, $TS(g)$.*

Proof. We construct a deterministic (conditional on θ) decision rule that use the realization of the state of one of the parties as the common randomization device.

Let $\mathcal{J}_i(g) := \{j \in \mathcal{N}\backslash\{i\} : (i, j) \in \bigcup_{k\in\mathcal{K}} \mathcal{Q}_k(g)\}$ denote the set of targets party i must match across all projects. If $\mathcal{J}_i(g) = \varnothing$, the result is immediate. If $i \notin \mathrm{Arm}(g)$, the result is also immediate: all payoff-relevant non-self tasks are performed by disarmed parties and can be implemented by full revelation. Hence suppose $\mathcal{J}_i(g) \neq \varnothing$ and $i \in \mathrm{Arm}(g)$. Choose a bottleneck target $r \in \arg\min_{j\in\mathcal{J}_i(g)} \tau_{ij}^{\mathrm{safe}}(g)$. For each $j \in \mathcal{J}_i(g)$, let $\tau_j := \tau_{ij}^{\mathrm{safe}}(g)$ and $L_j := 2\delta_j/\tau_j$.

Bottleneck partition. Apply the construction in the proof of Proposition 3 to the task (i, r). This yields distinct intended actions $x_r^1, \ldots, x_r^{M_r} \in \mathbb{T}$, success-zone endpoints $0 = \nu^0 < \nu^1 < \cdots < \nu^{M_r} = \tau_r$, and the associated success and failure intervals

$$S_r^m = \begin{cases} [\nu^{m-1}, \nu^m), & m = 1, \ldots, M_r - 1, \\ [\nu^{M_r-1}, \nu^{M_r}], & m = M_r, \end{cases}$$

$$F_r^1 = (\tau_r, \phi(\nu^1)), \quad F_r^{M_r} = [\phi(\nu^{M_r-1}), 1), \quad F_r^m = \phi(S_r^m) \quad \text{for } m = 2, \ldots, M_r - 1,$$

where $\phi(y) := \tau_r + \frac{1-\tau_r}{\tau_r} \cdot y$. Let $C_r^m := S_r^m \cup F_r^m$. By construction, the probability that $\theta \in S_r^m$ conditional on being in C_r^m is equal to τ_r, that is, $\frac{|S_r^m|}{|C_r^m|} = \tau_r$. In addition, following the recommendation x_r^m ensures that any state $\theta_r \in S_r^m$ is within a δ_r-neighborhood of x_r^m, that is, $S_r^m \subseteq B(x_r^m, \delta_r)$. Finally, there is no other action $y \in \mathbb{T}$ that matches the state θ_r with probability higher that τ_r, that is,

$$\sup_{y \in \mathbb{T}} \frac{|B(y, \delta_r) \cap C_r^m|}{|C_r^m|} \leqslant \tau_r.$$

Coordination Device. We next construct a variable that will play the role of the common randomization device based on the realization of θ_r. Let $L_m^s := \inf S_r^m = \nu^{m-1}$ and $L_m^f := \inf F_r^m$. For each m, define $\eta_m : C_r^m \to [0, 1)$ by

$$\eta_m(y) := \begin{cases} \dfrac{1}{2} - \dfrac{\tau_r}{2} + \dfrac{y - L_m^s}{|C_r^m|}, & y \in S_r^m, \\ \left(\dfrac{1}{2} + \dfrac{\tau_r}{2} + \dfrac{y - L_m^f}{|C_r^m|}\right) \pmod 1, & y \in F_r^m. \end{cases}$$

The map η_m is measure-preserving with respect to the normalized Lebesgue measure on C_r^m: on each piece it is affine with slope $1/|C_r^m|$, and $|S_r^m|/|C_r^m| = \tau_r$. Endpoint conventions for S_r^m and F_r^m affect η_m only on a Lebesgue-null set and do not affect the probability calculations that follow.

Deterministic rule. Define decision rule $\bar{\sigma}^g$ as follows. For every self-task (ℓ, ℓ), set $x_{\ell\ell} = \bar{x}_{\ell\ell} = \theta_\ell$. For every task (ℓ, j) with $\ell \notin \text{Arm}(g)$, set $x_{\ell j} = \bar{x}_{\ell j} = \theta_j$. For the task (i, r), if $\theta_r \in C_r^m$, set $x_{ir} = \bar{x}_{ir} = x_r^m$. For the tasks (i, j) with $j \in \mathcal{J}_i(g) \backslash \{r\}$, set

$$x_{ij} = \bar{x}_{ij} := \theta_j + L_j \left(\frac{1}{2} - \eta_m(\theta_r)\right) \pmod 1.$$

All expropriation actions are set to $\varnothing$. Remaining action components are set to arbitrary constants.

Party i's posterior. Conditional on the recommendation $x_{ir} = x_r^m$, the state θ_r is uniform on C_r^m, so $\eta_m(\theta_r)$ is uniform on $[0, 1)$. For $j \in \mathcal{J}_i(g) \backslash \{r\}$, x_{ij} is a translation of θ_j by a function of $\eta_m(\theta_r)$. Since θ_j is independent of θ_r and uniformly distributed, x_{ij} is independent of $\eta_m(\theta_r)$. Thus, the vector of non-bottleneck recommendations does not reveal additional information about $\eta_m(\theta_r)$ or θ_r, and $\eta_m(\theta_r)$ remains uniform conditional on i's information $I_i = (\theta_i, a_i)$.

For each $j \in \mathcal{J}_i(g) \backslash \{r\}$, the posterior of $\theta_j \mid I_i$ is uniform on the arc of length L_j centered at x_{ij}. The posterior of $\theta_r \mid I_i$ is uniform on C_r^m.

No expropriation. For every relevant target $j \in \mathcal{J}_i(g)$, party i cannot match θ_j with probability higher than $\tau_{ij}^{\text{safe}}(g)$. For irrelevant targets, party i receives only constant recommendations, and therefore it does not update its prior beliefs. Thus, refraining from expropriation is weakly optimal for all targets. Any other armed party receives only self-task or constant recommendations, so uninformed expropriation is unprofitable.

Project success. Task (i, r) is completed if and only if $\theta_r \in S_r^m$, equivalently $|\eta_m(\theta_r) - \frac{1}{2}| \leqslant \tau_r/2$. For $j \in \mathcal{J}_i(g)\backslash\{r\}$, task (i, j) is completed if and only if $d(x_{ij}, \theta_j) \leqslant \delta_j$, equivalently $|\eta_m(\theta_r) - \frac{1}{2}| \leqslant \tau_j/2$. Because $\tau_r \leqslant \tau_j$ for all $j \in \mathcal{J}_i(g)$, the completion events are nested: whenever task (i, r) is completed, every other task (i, j) with $j \in \mathcal{J}_i(g)$ is also completed. Let $\mathcal{J}_{i,k}(g) := \{j \in \mathcal{J}_i(g) : (i, j) \in \mathcal{Q}_k(g)\}$. If $\mathcal{J}_{i,k}(g) = \varnothing$, every task in project k is completed with probability one. Otherwise,

$$\mathbb{P}_{\bar{\psi}}(E_k) = \min_{j \in \mathcal{J}_{i,k}(g)} \tau_j = \min_{(\ell,j) \in \mathcal{Q}_k(g)} \tau_{\ell j}^{\text{safe}}(g),$$

where $\bar{\psi}(a, \theta) = p(\theta)\bar{\sigma}^g(a \mid \theta)$. Moreover, no alternative cooperative action can ensure task (i, j) is completed with probability greater than τ_j. Therefore, for any deviation in cooperative actions, the probability that project k succeeds is bounded above by $\min_{j \in \mathcal{J}_{i,k}(g)} \tau_j$, which is the probability attained by the recommended cooperative actions. Since project values are nonnegative, the prescribed cooperative actions are obedient.

Total surplus. Summing over projects,

$$\mathbb{E}_{\bar{\psi}}[TS(a, g; \theta)] = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(\ell,j) \in \mathcal{Q}_k(g)} \tau_{\ell j}^{\text{safe}}(g),$$

which matches the upper bound established for the stochastic decision rule. □

A.6 Proof of Proposition 4

Proof. (i) Because ϕ^c is injective, we can choose an ownership profile g such that $g^c(j) = j$ for all $j \in \mathcal{N}$. Since $\bar{\mathcal{R}}_k = \varnothing$, each project has tasks of the form $\mathcal{Q}_k(g) = \{(j, j) : j \in \mathcal{R}_k\}$. Hence, $\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = 1$ for every k and therefore $TS(g) = V_{\text{total}}$.

(ii) Because ϕ^e is injective, choose g such that $g^e(j) = j$ for all $j \in \mathcal{N}$. Since $\mathcal{E} = \mathcal{N}$, no party can expropriate information of any other party, so $TS(g) = V_{\text{total}}$. □

A.7 Optimal Decision Rule Under Arbitrary Expropriation Costs

This appendix extends the characterization of an optimal decision rule to the case of arbitrary expropriation costs, under the assumption that only one armed party receives information about others.

Let $\mathcal{J}_i(g) := \{j \in \mathcal{N}\backslash\{i\} : (i, j) \in \bigcup_{k \in \mathcal{K}} \mathcal{Q}_k(g)\}$ denote the set of targets whose states the receiver must match, and let $\mathcal{J}_{i,k}(g) := \{j \in \mathcal{J}_i(g) : (i,j) \in \mathcal{Q}_k(g)\}$ denote the subset required for project k.

For each target $j \in \mathcal{J}_i(g)$, let q_j denote the probability that party i receives a correct cooperative-action recommendation $x_{ij} = \bar{x}_{ij} = \theta_j$. Denote by $R \subseteq \mathcal{J}_i(g)$ the set of targets the party i is recommended to expropriate. Define the contribution to the total surplus of an expropriation attempt that succeeds with probability q:

$$e_{ij}(q) := q(b_{ij} - \lambda_{ij}) - (1-q)c_{ij}.$$

Define the following value function:

$$\begin{aligned} W_R(g) := \max_{\{q_j\},\{m_k\}} \quad & \sum_{k\in\mathcal{K}} V_k m_k + \sum_{j\in R} e_{ij}(q_j) \\ \text{subject to} \quad & q_j \in [\tau^\star_{ij}, 1] \quad \text{for all } j \in R, \\ & q_j \in [1/2, \tau^\star_{ij}] \quad \text{for all } j \in \mathcal{J}_i(g)\backslash R, \\ & 0 \leqslant m_k \leqslant 1 \quad \text{for all } k \in \mathcal{K}, \\ & m_k \leqslant q_j \quad \text{for all } k \in \mathcal{K} \text{ and } j \in \mathcal{J}_{i,k}(g). \end{aligned} \tag{A.24}$$

When $\mathcal{J}_{i,k}(g) = \varnothing$, the constraint $m_k \leqslant q_j$ is absent, and $m_k = 1$ at the optimum.

Proposition 8 (Arbitrary expropriation costs with a single armed receiver)**.** *Fix an owner $g \in \mathcal{N}$. Suppose there exists an armed party $i \in \mathcal{N}$ such that every task in $\bigcup_{k\in\mathcal{K}} \mathcal{Q}_k(g)$ is one of the following: (i) a self-task (ℓ,ℓ), (ii) a task (ℓ,j) performed by a disarmed party $\ell \notin \mathrm{Arm}(g)$, (iii) a task (i,j). Then the maximum attainable total surplus is*

$$TS(g) = \max_{R\subseteq\mathcal{J}_i(g)} W_R(g).$$

Proof. We first establish the upper bound and then construct an obedient decision rule that attains it.

Upper bound. Fix any obedient decision rule σ and the associated joint distribution $\psi(a,\theta) := p(\theta)\sigma(a \mid \theta)$. Consider an on-path information set of the party i.

Obedience in expropriation implies: if $j \in R$, then $q_j \geqslant \tau^\star_{ij}$; if $j \notin R$, then $q_j \leqslant \tau^\star_{ij}$. Hence, $q_j \in [\tau^\star_{ij}, 1]$ for $j \in R$ and $q_j \in [1/2, \tau^\star_{ij}]$ for $j \notin R$.

Project k succeeds only if party i completes every task (i,j) with $j \in \mathcal{J}_{i,k}(g)$, and each such task is completed with probability at most q_j. Therefore, the project-k success probability m_k satisfies $m_k \leqslant \min_{j\in\mathcal{J}_{i,k}(g)} q_j$, with $\min\varnothing := 1$. The contribution of expropriation attempts to total surplus is at most $\sum_{j\in R} e_{ij}(q_j)$.

The total surplus at this information set is thus bounded above by the value of the program (A.24) defining $W_R(g)$. Taking expectations over information sets and maximizing over R,

$$\mathbb{E}_\psi[TS(a, g; \theta)] \leqslant \max_{R \subseteq \mathcal{J}_i(g)} W_R(g).$$

Attainability. Let $R^* \in \arg\max_R W_R(g)$ and let (q^*, m^*) solve (A.24) at R^*. Because the objective is increasing in each m_k, the optimum sets m_k to its upper bound:

$$m_k^* = \min_{j \in \mathcal{J}_{i,k}(g)} q_j^*,$$

with $m_k^* = 1$ when $\mathcal{J}_{i,k}(g) = \varnothing$.

Let $\xi \sim \mathrm{Unif}[0, 1]$ be a common randomization device. The mediator recommends the following cooperative actions:

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j & \text{if } \xi \leqslant q_j^*, \\ -\theta_j & \text{if } \xi > q_j^*, \end{cases} \qquad j \in \mathcal{J}_i(g).$$

For $j \in R^*$, the mediator recommends $\zeta_{ij} = x_{ij}$; for $j \notin R^*$, it recommends $\zeta_{ij} = 0$. All self-tasks and tasks performed by disarmed parties are set equal to the true states and are completed with probability one. Payoff-irrelevant cooperative actions are set arbitrarily.

Posteriors and obedience. Conditional on its recommendation, party i cannot match θ_j with probability higher than q_j^*. Thus for $j \in R^*$, $q_j^* \geqslant \tau_{ij}^\star$, and the recommended expropriation attempt is weakly optimal. For $j \notin R^*$, $q_j^* \leqslant \tau_{ij}^\star$, and no expropriation is weakly optimal.

The cooperative actions are also obedient. The recommended action for each target are more likely to match the state, so following it is weakly optimal task-by-task. Moreover, because all targets use the same ξ, task completions are nested: task (i, j) is completed if and only if $\xi \leqslant q_j^*$. Hence, project k succeeds with probability $\min_{j \in \mathcal{J}_{i,k}(g)} q_j^* = m_k^*$ when $\mathcal{J}_{i,k}(g) \neq \varnothing$, and with probability one otherwise. No deviation in cooperative actions can raise any task's completion probability beyond q_j^*, so project-k success cannot exceed m_k^* under any deviation. Since project values are nonnegative, the prescribed cooperative actions are obedient.

Total surplus. The total surplus under this rule equals

$$\sum_{k \in \mathcal{K}} V_k m_k^* + \sum_{j \in R^*} e_{ij}(q_j^*) = W_{R^*}(g) = \max_{R \subseteq \mathcal{J}_i(g)} W_R(g),$$

matching the upper bound. □